

TEST - 9010
(EXPLANATION & SOURCE)

1. (c)

Explanation:

Karewa is a plateau-like landforms which are 13000-18000-metre-thick deposits of alluvial soil and sediments like sandstone and mudstone that remain tucked away in the folds of the surrounding mountains, particularly the Pir Panjal range of the Himalayas that borders the Kashmir valley on the southwest.

Statement 1 is correct and 2 is incorrect:

Kashmir's highly fertile alluvial soil deposits called 'karewas'.

The fertility of these patches is believed to be the result of their long history of formation. It was formed **during the Pleistocene period (2.6 million years to 11,700 years ago)**, the Pir Panjal range blocked the natural drainage in the region and formed a lake spanning 5,000 sq km (roughly three times the size of Delhi).

Statement 3 is correct: This makes it ideal for the growth of crops like **saffron, almonds, apples and several other cash crops.**

X Factor:

- The Srinagar airport is built on the **Damodar karewa** in Budgam.
- Kashmir Saffron received Geographical **Indication (GI) tag in 2020** for its longer and thicker stigmas, deep-red colour, high aroma and bitter flavour.
- The karewa sediments hold fossils and remnants of many human civilisations and habitations.

Source: NCERT Class 11, India Physical Environment Geography Chapter 2, Structure and Physiography

<https://www.downtoearth.org.in/news/environment/heritage-erased-how-the-kashmir-valley-s-ancient-mound-formations-are-being-levelled-81683>

2. (a)

Explanation:**Option (a) is correct:**

The post-drift studies provided considerable information that was not available at the

time Wegener put forth his concept of continental drift. **Sea Floor Spreading** is one the most prominent, which is based on the following facts:

- It was realised that all along the **mid oceanic ridges**, volcanic eruptions are common and they bring huge amounts of lava to the surface in this area.
- The rocks equidistant on either side of the crest of mid-oceanic ridges show remarkable similarities in terms of period of formation, chemical compositions and **magnetic properties. Rocks closer to the mid-oceanic ridges have normal polarity and are the youngest. The age of the rocks increases as one moves away from the crest.**
- **The ocean crust rocks are much younger than the continental rocks.** The age of rocks in the oceanic crust is nowhere more than 200 million years old. Some of the continental rock formations are as old as 3,200 million years.
- **The sediments on the ocean floor are unexpectedly very thin.** Scientists were expecting, if the ocean floors were as old as the continent, to have a **complete sequence of sediments for a period of much longer duration.** However, nowhere was the sediment column found to be older than 200 million years.
- The deep trenches have deep-seated earthquake occurrences while in the mid-oceanic ridge areas, the quake foci have shallow depths.

X-Factor:

Continental Drift Theory: By Alfred Wegner in 1912. This was regarding the distribution of the oceans and the continents.

- According to Wegener, all the continents formed a single continental mass and Mega Ocean surrounded the same.
- The super continent was named **PANGAEA**, which meant all earth. The mega-ocean was called **PANTHALASSA**, meaning all water.
- He argued that, around 200 million years ago, the super continent, Pangaea, began

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to split. Pangaea first broke into two large continental masses as **Laurasia** and **Gondwanaland** forming the northern and southern components respectively.

- Subsequently, Laurasia and Gondwanaland continued to break into various smaller continents that exist today.

Source: NCERT Class 11th Fundamentals of physical geography, Chapter 4, Page no. 30-33

3. (d)

Explanation:

Landslides are relatively rapid and perceptible movements. The materials involved are relatively dry. The size and shape of the detached mass depends on the nature of discontinuities in the rock, the degree of weathering and the steepness of the slope. Depending upon the type of movement of materials several types are identified in this category.

Statement 1 is incorrect: Slump is slipping of one or several units of rock debris with a backward rotation with respect to the slope over which the movement takes place.

Statement 2 is incorrect and statement 3 is correct: Rapid rolling or sliding of earth debris without backward rotation of mass is known as debris slide. Debris fall is nearly a free fall of earth debris from a vertical or overhanging face. **Sliding of individual rock masses down bedding, joint or fault surfaces is rockslide.** Over steep slopes, rock sliding is very fast and destructive. Slides occur as planar failures along discontinuities like bedding planes that dip steeply.

Statement 4 is correct: Creep is the slow downslope movement of material under gravity. It generally occurs over large areas.

X-Factor:

- Rock fall is free falling of rock blocks over any steep slope keeping itself away from the slope.

Three types of creep occur:

- seasonal movement or creep within the soil – due to seasonal changes in soil moisture and temperature, e.g. frost heave processes

- continuous creep – where the soil and rock debris isn't strong enough to resist gravity
- progressive creep – where the soil and rock debris suddenly reaches a critical state such that it will fail due to other factors

Source: NCERT Class 11th Fundamentals of physical geography, Chapter 6, Page no. 51-52

<https://www.nature.scot/landforms-and-geology/scotlands-rocks-landforms-and-soils/landforms/landslides/types-landslide>

4. (b)

Explanation:

Incised or Entrenched Meanders In streams that flow rapidly over steep gradients, normally erosion is concentrated on the bottom of the stream channel. Also, in the case of steep gradient streams, lateral erosion on the sides of the valleys is not much when compared to the streams flowing on low and gentle slopes.

Statement 1 is correct and Statement 2 is incorrect: Because of active lateral erosion, streams flowing over gentle slopes, develop sinuous or meandering courses. It is common to find meandering courses over floodplains and delta plains where stream gradients are very gentle.

Statement 3 is incorrect:

Incised meander associated with Rejuvenation: But very deep and wide meanders can also be found cut in hard rocks. Such meanders are called incised or entrenched meanders. Meander loops develop over original gentle surfaces in the initial stages of development of streams and the same loops get entrenched into the rocks normally due to erosion or slow, continued uplift of the land over which they start. They widen and deepen over time and can be found as **deep gorges** and canyons in hard rock areas. They give an indication on the status of original land surfaces over which streams have developed.

X-factor:

Rejuvenation:

Over a long time, a river forms a generally smooth profile. In terms of transporting water and sediment it is the most efficient

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profile to have. Rivers that have this form of profile are in balance with the environment (equilibrium). Typically, a river will always try to form a concave shaped profile.

From time to time something can occur that changes and de-stabilises this equilibrium. This causes the river to vertically erode its channel to re-establish the its long, smooth concave profile. This renewed period is known as rejuvenation.

Source: Ncert, physical geography page no. 60

<https://www.alevelgeography.com/rejuvenation/>

5. (d)

Explanation:

Statement 1 is correct: The Shendurney Wildlife Sanctuary is a **tropical semi-evergreen forest** and is home to animals like elephants, leopards, bison, sambar, wild boar, Lion-tailed macaques, langurs.

Statement 2 is correct: Shendurney wildlife sanctuary, **part of Agsthyamali Biosphere Reserve**, owes its name to the endemic species, *Ghuta Travancorica*, locally known as 'Chenkurinji'. The sanctuary is a valley of green splendour acclaimed for its rich biodiversity lying on either side of the Shendurney river.

Statement 3 is correct: Located on the north of Kulathupuzha Valley and separated by the Churuttumala Ridge, it is bounded by the Sahyadri range on the east. Most of the Sanctuary area is hilly and interspersed with ravines. There are quite a few rugged and steep peaks, and the Sanctuary is, therefore, inaccessible in many places. Before the advent of the British, the entire region was covered with thick evergreen forests of great timber value. White planters, who took possession of large tracts ostensibly for coffee and tea cultivation, ruthlessly destroyed these forests for timber. Most of the areas have since been reclaimed and brought under forest cover. **There is no tribal settlement inside the Sanctuary.**

Statement 4 is correct: Shendurney Wildlife Sanctuary, **the name is said to have been derived from an endemic plant, Glutatravancoria, locally known as Chenkurinji**, lies on either side of Shendurney river.

X-Factor:

- The Agasthyamala Biosphere Reserve located in the Western Ghats in the south of the country consisting mostly of tropical forest. It is the site home to 2,254 species of higher plants including about 400 that are endemic.
- Three wildlife sanctuaries, Shendurney, Peppara and Neyyar, are located in the site, as well as the Kalakad Mundanthurai Tiger reserve.
- The reserve is home to Kani tribes from both Tamil Nadu and Kerala. Its prominence in the epic Ramayana has made it a famous site for Hindu pilgrimages.

Source:

<https://forest.kerala.gov.in/index.php/wildlife/2015-03-16-09-50-24/2015-06-26-09-04-29/shendurney-wildlife-sanctuary><https://www.keralatourism.org/ecotourism/destinations/thenmala-shendurney><http://datazone.birdlife.org/site/factsheet/shendurney-wildlife-sanctuary-iba-india>

<https://en.unesco.org/biosphere/aspac/agasthyamala>

6. (c)

Explanation:

Statement 1 is correct: Karst is a type of landscape where the dissolving of the bedrock has created sinkholes, sinking streams, caves, springs, and other characteristic features. Karst is associated with soluble rock types such as limestone, marble, and gypsum. Any limestone or dolomitic region showing typical landforms produced by the action of groundwater through the processes of solution and deposition is called Karst topography after the typical topography developed in limestone rocks of Karst region in the **Balkans adjacent to Adriatic Sea.**

Statement 2 is correct: **Stalactites, Stalagmites and Pillars are the depositional landforms of Karst topography.** Stalactites hang as icicles of different diameters. Normally they are broad at their bases and taper towards the free ends showing up in a variety of forms. Stalagmites rise up from the floor of the caves. In fact, stalagmites form due to

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dripping water from the surface or through the thin pipe, of the stalactite, immediately below it. The stalagmite and stalactites eventually fuse to give rise to columns and pillars of different diameters.

X-Factor:

Erosional landforms of karst topography:

- Small to medium sized round to sub-rounded shallow depressions called **swallow holes** form on the surface of limestones through solution.
- A **sinkhole** is an opening more or less circular at the top and funnel-shaped towards the bottom with sizes varying in area.
- When sink holes and dolines join together because of slumping of materials along their margins or due to roof collapse of caves, long, narrow to wide trenches called **valley sinks or Uvalas form**.
- Gradually, most of the surface of the limestone is eaten away by these pits and trenches, leaving it extremely irregular with a maze of points, **grooves and ridges or lapies**.

Source: NCERT Class 11th Fundamentals of physical geography, Chapter 7, Page no. 60-62

<https://www.nps.gov/subjects/caves/karst-landscapes.htm#:~:text=Karst%20is%20a%20type%20of,limestone%2C%20marble%2C%20and%20gypsum.>

7. (a)

Explanation:

Statement 1 is correct: In Western countries, males outnumber females in rural areas and females outnumber the males in urban areas. In developing countries like Nepal, Pakistan and India the case is reverse. According to World Bank, Urban population, female (% of total) in India was reported at 15.5 % in 2015.

Statement 2 and 3 are incorrect: Population ageing is the process by which the share of the older population becomes proportionally larger. This is a new phenomenon of the twentieth century. **In most of the developed countries of the world, population in higher age groups has increased due to increased life**

expectancy. With a reduction in birth rates, the proportion of children in the population has declined. Developed countries have low death due to advanced medical facilities and increased life expectancies.

Currently, **India is going through demographic transition, where its young population outnumbers the ageing population.** Over half of the Indian population is below age of 30.

X-Factor:

- In regions where gender discrimination is rampant, the sex ratio is bound to be unfavourable to women.
- In India – literacy rate denotes the percentage of population above 7 years of age, who is able to read, write and have the ability to do arithmetic calculations with understanding.
- Sex ratio in Asian urban areas remains male dominated due to the predominance of male migration.

Source: NCERT 12th: Fundamentals of Human Geography, Chapter 3, Page no. 19

<https://indianexpress.com/article/explained/half-indias-population-under-age-30-nfhs-explained-7910458/>

[https://tradingeconomics.com/india/urban-population-female-percent-of-total-wb-data.html#:~:text=Urban%20population%2C%20female%20\(%25%20of%20total\)%20in%20India%20was%20reported,compiled%20from%20officially%20recognized%20sources.](https://tradingeconomics.com/india/urban-population-female-percent-of-total-wb-data.html#:~:text=Urban%20population%2C%20female%20(%25%20of%20total)%20in%20India%20was%20reported,compiled%20from%20officially%20recognized%20sources.)

8. (c)

Explanation:

A massive zone of cool water in the North Atlantic Ocean is slowing the melting of Iceland's glaciers but since 2011 the speed of Iceland's melting slowed, resulting in about half as much ice loss annually.

Statement 1 is correct: Blue Bob is a patch of unusually cold water located south of Iceland and Greenland. But not much is known about it, and scientists are still investigating the impact of this cold-water area on surrounding regions. Blue Blob has been linked to **cooler air temperatures over Iceland's glaciers and cooler waters.**

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Statement 2 is incorrect: The **North Atlantic Oscillation** is a **prominent mode of variability with important impacts from the polar to subtropical Atlantic and surrounding landmasses**. It is most pronounced during boreal winter. It is associated with changes in wind strength and direction, heat and moisture transport, and the frequency and strength of storms. **Hence it is not associated with slow melting of glaciers rather it will fasten the melting process.**

Statement 3 is correct: Before the **Blue Blob**, a long-term cooling trend in the same region, called the **Atlantic Warming Hole**, reduced sea surface temperatures by about **0.4 to 0.8 degrees Celsius** during the last century and may continue to cool the region in the future. The Warming Hole is a **slowdown of the Atlantic Meridional Overturning Circulation (AMOC)**. AMOC is an ocean current that brings warm water up from the tropics to the Arctic, thus reducing the amount of heat delivered to the region.

X Factor:

Since 2013, India has had observer status in the Arctic Council, which is the predominant inter-governmental forum for cooperation on the environmental and development aspects of the Arctic.

Source: <https://telanganatoday.com/blue-blob-is-slowng-glacier-melting>

<https://weather.com/en-IN/india/science/news/2022-02-23-here-how-mysterious-blue-blob-could-slow-down-glacial-melting>

9. (b)

Explanation:

Recently Union Cabinet announced its approval of supply of fortified rice across three phases throughout the Targeted Public Distribution System (TPDS) under the National Food Security Act (NFSA), Integrated Child Development Services (ICDS), Pradhan Mantri Poshan Shakti Nirman-PM POSHAN erstwhile Mid-Day Meal Scheme (MDM)] and Other Welfare Schemes (OWS) of Centre and state government by 2024.

Statement 1 is incorrect: The Food Safety and Standards Authority of India (FSSAI) defines fortification as “deliberately

increasing the content of essential micronutrients in a food so as to improve the nutritional quality of food and to provide public health benefit with minimal risk to health”. (Addition of key vitamins and minerals such as iron, iodine, zinc, Vitamins A and D to staple foods such as rice, wheat, oil, milk and salt to improve nutritional value and provide a public health benefit with minimal risk to health.)

Fortification does not require any changes in the eating patterns or food habits of people

Statement 2 is correct: Various technologies are available for **rice fortification**, such as coating and dusting. For rice fortification in India, **‘extrusion’** is considered to be the best technology. This involves the production of fortified rice kernels (FRKs) from a mixture using an extruder machine.

Statement 3 is incorrect: Fortified rice is **nearly identical to traditional rice in aroma, taste, and texture.**

X factor

- Rice fortification ecosystem has been enhanced significantly on boarding rice millers, FRK manufacturers, industries and other stakeholders for production and supply of fortified rice to meet the target requirement.
- Fortification of rice is found to be a cost-effective and complementary strategy to increase vitamin and mineral content in diets which helps in fighting anaemia and malnutrition in the country.
- Extrusion is a **process where a material undergoes plastic deformation by the application of a force causing that material to flow through an orifice or die.**

Source:

<https://www.thehindu.com/news/national/india-rice-fortification-scheme-experts-concern-explained-food/article65453762.ece>

<https://pib.gov.in/PressReleaseIframePage.aspx?PRID=1850981>

<https://indianexpress.com/article/india/cabinet-fortified-rice-cost-rs-2700-crore-per-year-7860261/>

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10. (d)

Explanation:

Coral reefs are unique and the most diverse marine ecosystems on Earth. They are believed to have existed for about 200 million years. It is estimated that it took corals about 50 million years to reach their present level of diversity.

10 per cent of coral reefs have degraded; another 30 per cent are likely to disappear within next 20 years across the globe. According to biologists, if proper conservation and management measures are not taken, all coral reefs of the Indo-Malayan region may disappear in the next 40 years.

Statement 1 ,2 and 3 are correct:

When the ocean water becomes too warm, the **corals bleach** as the symbiotic algae (Zooxanthallae) leave them. If the bleaching continues, **its growth is obstructed or it eventually dies**.

Acanthaster planci, also known as **Crown-of-Thorns Star fish**, eats corals during night. They may **destroy the entire coral reefs if found in huge numbers**.

Coral reefs are also affected by **coral diseases such as black band and white band** due infectious microorganisms introduced by the human population that live on the coastal regions.

X Factor:

Mangrove forests play a crucial role in helping the coral reef system by acting as filters and providing protection from cyclones, storms and tsunamis.

Destruction of mangrove forest in the reef areas should be prohibited. Planting of mangrove species in reef areas can be undertaken with the involvement and support of the local community.

Source

<https://www.downtoearth.org.in/blog/wildlife-biodiversity/marine-pollution-coastal-development-coral-reefs-need-to-be-saved-76298#:~:text=India%20has%20four%20coral%20reef,the%20humanity%20from%20natural%20calamities>

11. (a)

Explanation:

The Food Safety and Standards Authority of India (FSSAI) has introduced the Jaivik Bharat logo to help customers identify authentic organic food. Any product that has the Jaivik Bharat logo would have to meet specific standards of organic farming

Statement 1 is correct.: India ranks first in number of organic farmers and ninth in terms of area **under organic farming**. **Sikkim became the first State in the world to become fully organic and other States including Tripura and Uttarakhand have set similar targets**. North East India has traditionally been organic and the consumption of chemicals is far less than rest of the country. Similarly, the tribal and island territories are being nurtured to continue their organic story.

Statement 2 is correct: According to the organic regulations, from 1 July 2018 dealers were permitted to sell organic products only if the labels bore the Jaivik Bharat logo but this has now been extended to 01 April 2019. **Organic Foods are considered authentic organic only if they have Organic Production (NPOP) and Participatory Guarantee System for India (PGS-India) registration certificates**. Once the certificates are procured the dealers can then submit these certificates to procure permission to print the logo on the organic products. Besides the requirement of the Jaivik Bharat Logo, the organic foods need to also comply with the labelling and packaging requirements as prescribed in the Food Safety and Standards (Packaging and Labelling) Regulations, 2011.



Statement 3 is incorrect: The portal on Organic Food from India is a regulatory portal developed by **FSSAI, APEDA and (PGS-India)** on Organic food standards, certification processes, information relating

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to FBOs, their products and geographical areas in which they are available.

X Factor:

- The logo is supported with the tagline "Jaivik Bharat", at the bottom, which signifies Organic Food from India. Employing a simple approach to communicate an effective message.
- **The green color of the logo symbolizes nature along with articulating that these foods are produced in environment-friendly manner without using chemical fertilizers and pesticides.**
- It highlights the letter "O": To show a sturdy relationship with the same, leaves have also been used in the logo in the form of a tick mark incorporated inside the circle represented by the letter "O".
- The logo showcases that the product bearing it has been authenticated as organic for the choice for consumption.

Source:

<https://jaivikbharat.fssai.gov.in/>
<https://pib.gov.in/PressReleasePage.aspx?PRID=1645497>
<https://www.prakati.in/jaivik-bharat-certification-organic-food-india/>
<https://www.indiatimes.com/explainers/news/how-sikkim-became-worlds-first-organic-state72280.html#:~:text=This%20recognition%20implies%20that%20the,implement%20a%20100%25%20organic%20policy.&text=The%20citation%20reads%2C%20%E2%80%9CSikkim%20state,state%20with>

12. (d)

Explanation:

Recognized by UNESCO as one of the world's eight most important biodiversity hotspots, Western Ghats contains more than 30 percent of all species of plant, fish, reptile, amphibian, bird and mammal found in the whole country.

The Gadgil Commission was appointed under the chairmanship of ecologist Madhav Gadgil in 2010. This commission, also known as The Western Ghats Ecology Expert Panel (WGEEP), **was based on the**

environmental research of the entire Western Ghats of India.

Statement 2 is incorrect: The committee classified the extensive region of 64% of the Western Ghats, spanning over six States, 44 districts and 142 talukas, into Ecologically Sensitive Zones, called ESZ 1, ESZ 2 and ESZ 3. Suggested the need to desist from creating new hill stations, changing the land use from farmland to non-farm land and the stoppage of diversions of rivers in order to protect the ecology of the region. Further it suggested to completely stop developmental activities like mining, construction of thermal power plants, dams etc in ESZ-1.

Statement 1 is incorrect: It proposed a bottom to top approach instead of a top to bottom approach, basically, the decentralization and more powers to local authorities.

X-factor:

- The Western Ghats is a mountain range that runs along the western side of India. These Ghats are in fact older than the Himalayas.
- It is one of the eight hottest hotspots of biological diversity in the world.
- It originates near the border of Gujarat and Maharashtra, and runs through the states of Maharashtra, Goa, Karnataka, Tamil Nadu, and Kerala, finally ending at Kanyakumari.
- The Silent Valley National Park in Kerala, which forms part of these Ghats, is one among the last tracts of virgin tropical evergreen forest in India.
- *Strobilanthes kunthianus* is locally known as Kurinji or Neelakurinji. Its flowers bloom only once in 12 years. Only the shrub is known to grow in the Shola Forests of the Western Ghats.
- The range is known as Sahyadri in Maharashtra and Karnataka. Anaimudi (2695 m), is the highest peak of the Western Ghats, which is situated in Eravikulam National Park, Kerala. Mullayanagiri (1,950 m) is the highest peak in Karnataka. Doddabetta (2637 m), is the highest peak of Tamil Nadu.

Source:<https://www.indiatimes.com/explainers/news/gadgil-report-on-western-ghats-ecology-cop26-553094.html>

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<https://indianexpress.com/article/india/local-communities-no-role-economic-decisions-unscientific-kasturirangan-report-madhav-gadgil-8054612/>
<https://www.nativeplanet.com/travel-guide/compelling-facts-about-the-western-ghats-that-will-startle-you-005029.html>

13. (c)

Explanation:

Statement 1 is correct: Foot loose industries can be located in a wide variety of places. They are not dependent on any specific raw material, weight losing or otherwise.

Statement 2 is incorrect and statement 3 is correct: They largely depend on component parts which can be obtained anywhere. They produce in small quantity and also employ a small labour force. These are generally not polluting industries. The important factor in their location is accessibility by road network.

X-factor:

Industries based on Ownership:

(a) Public Sector Industries are owned and managed by governments. In India, there were a number of Public Sector Undertakings (PSUs). Socialist countries have many state owned industries. Mixed economies have both Public and Private sector enterprises.

(b) Private Sector Industries are owned by individual investors. These are managed by private organisations. In capitalist countries, industries are generally owned privately.

(c) Joint Sector Industries are managed by joint stock companies or sometimes the private and public sectors together establish and manage the industries.

Source: Human Geography 12th NCERT; Chapter-6; Page- 51

14. (b)

Explanation:

The ability of the air to hold water vapour **depends entirely on its temperature**. The percentage of moisture present in the atmosphere as compared to its full capacity at a given temperature is known as the

relative humidity. With the change of air temperature, the capacity to retain moisture increases or decreases and the relative humidity is also affected. It is greater over the oceans and least over the continents

Distribution of relative humidity:

Statement 1 is incorrect: The horizontal distribution of relative humidity on earth's surface is zonal in **character**. **Equatorial regions are characterized by highest relative humidity. It gradually decreases towards subtropical high-pressure belts, where it becomes minimum** (between 25 degrees to 35-degree latitudes). It further increases poleward.

Statement 2 is correct: The zones of high and low relative humidity shift northward and southward with northward and southward migration of the sun respectively. Seasonal distribution of relative humidity is largely controlled by latitudes. Maximum relative humidity is found during summer season between 30-degree N and 30-degree S latitudes while high latitudes record relative humidity more than average value during winters.

Statement 3 is correct: There is maximum relative humidity in the morning whereas lowest value is recorded in the evening.

X-factor:

Specific humidity: The actual amount of the water vapour present in the atmosphere is known as the absolute humidity. It is the weight of water vapour per unit volume of air and is expressed in terms of grams per cubic metre.

The absolute humidity differs from place to place on the surface of the earth.

Source: NCERT, fundamental of physical geography, Page no.99

Physical geography by Savinder singh Page no. 457

15. (d)

Explanation:

Pair 1 is correct: Gour or Mushroom rock are formed by sand blasting effect of wind due to which grooves and hollows are cut in the rock surfaces, carving them into rock pedestals, which are further eroded **near their bases where the friction is greatest**.

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Pair 2 is correct: **Zeugen** are tabular masses which have a layer of soft rocks lying beneath **resistant rocks, formed into weird -looking 'ridge and furrow landscape' due to wind abrasion.**

Pair 3 is correct: **Dreikanter** are the type of **ventifacts with three wind-faceted surfaces.** Ventifacts have flat facets with sharp edges formed by sand blasting.

Pair 4 is correct: **Inselberg** are **isolated residual hills** rising abruptly from the level ground, characterised by their very steep slopes and rather rounded tops.

X Factor:

In Libya and Egypt stony desert are termed as serir and in Africa stony desert are called as reg.

In Turkestan, sandy deserts are known as Koum.

Source: GC Leong Chapter 7, Arid and Desert landforms Page 56,57,58,59

16. (b)

Explanation:

Coral reefs are unique and the most diverse marine ecosystems on Earth. They play a crucial role in supporting the flora and fauna in the marine ecosystem.

Statement 2 is correct: The “coral modules” are manmade structures with 2,000 living coral attached. **They were designed to help reverse the effects of climate change, which has killed almost all the islands' coral reefs.**

Statement 1 is incorrect: According to the Global Coral Reef Monitoring Network between 2010 and 2020, rising global temperatures **killed 14 percent of the world's coral reefs.**

Statement 3 is incorrect: The “Coral modules” were created by climate scientist and marine biologist Deborah Brosnan and entrepreneur John Paul DeJoria with the help of **the Antigua and Barbuda government. The model has not been introduced yet in India.**

X Factor:

The Ocean Shot project led by scientist Dr. Deborah Brosnan in partnership with global philanthropist and entrepreneur John Paul DeJoria is the first of its kind to take action for the oceans.

The initiative plans to use technology that mimics the design and shape of natural reefs to provide opportunities for colonization by coral and other marine life. The constructed reef modules will also help protect nearby coastal communities from storm surge and sea-level rise.

Source:

https://news.yahoo.com/manmade-coral-reef-deployed-off-214500920.html?guccounter=1&guce_referrer=aHR0cHM6Ly93d3cuZ29vZ2xlLmNvbS8&guce_referrer_sig=AQAAEn2CiXfPQiQowQ_ZJmFzOJ4MB-6H8gduPdCgXez-IsRuwDm6KheXpd0XUW0bhBJYr0QLyV4lqePZC1emb0iR-J1jTy1Ruc_4NzeTrP3-uUQCs3Cc9tIMX2HlOp8HhWGxO7ueMSLdTX7oU-hYjrrNAVRuMkbbs3zxZsj4rgRJ3L
<https://www.globalcitizenforum.org/ocean-shot/>
<https://thehill.com/changing-america/sustainability/climate-change/3674117-new-manmade-coral-reef-deployed-off-coast-of-islands-in-hopes-of-restoring-ocean-ecosystem/>

17. (b)

Explanation:

Aeolian landforms:

Depositional landforms: Sand dunes:

Statement 1 is incorrect: Barchan are **crescentic or moon shaped dunes which occur individually or in groups.**

Statement 2 is correct: Barchans arise in deserts where the ground is **hard and flat and strong winds blow sand in one direction.** Thousands of the dunes can occupy a narrow strip. Fueled by wind, the dunes travel at different paces and can traverse up to 100 meters in a year.

Statement 3 is incorrect: Barchans **windward side is convex and gently -sloping while the leeward side, being sheltered, is concave and steep.**

The collisions prevent dunes from growing out of control. If two barchan dunes collide, they can merge into one crescent or they can split up into multiple smaller barchans.

X Factor:

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- Scientists have observed both collisions and calving in barchan fields.
- Large longitudinal sand dunes found abundantly in the western part of Great Indian desert.

Source:

<https://www.sciencenews.org/article/breakups-maintain-barchan-dune-fields-somehow>

GC Leong book, Chapter 7, Arid and Desert landforms Page no. 59

18. (c)

Explanation:

Statement 1 is incorrect: Borra caves are also called Borra Guhalu and are **located on the East Coast of India**. The caves are one of the largest in India at an elevation of about seven hundred and five meters.

Statement 2 is correct: million-years-old Borra Caves and volcanic ash deposits said to have **originated from the volcanic eruption of Toba in Indonesia** 73,000 years ago.

Statement 3 and 4 are correct: Due to the water activity cutting through rich limestone, **many stalagmites and stalactites have formed over the years**. According to researchers, the cave is highly valuable for anthropological research, as excavations carried out earlier unearthed stone tools of **middle Paleolithic culture dating between 30,000 and 50,000 years ago**. These findings **confirm human habitation** in that area.

X-Factor:

Borra Caves is situated in the crown of **Ananthagiri hills of Araku Valley**. These limestone caves are a million-year-old and the largest caves in India at an elevation of about 705 m. The Borra Caves were formed **due to the flow of Gosthani River on limestone deposits**. The caves have impressive stalactite and stalagmite formations of Shiva-Parvati, mother-child, crocodile, human brain, *rishi's* beard and cow's udder among others.

Source:

<https://timesofindia.indiatimes.com/travel/destinations/what-makes-million-year-old-borra-caves-near-visakhapatnam-so-unique/articleshow/74337174.cms>

<https://www.thehindu.com/news/national/andhra-pradesh/in-frames-the-ancient-caves-of-borra/article65848443.ece/photo/1/>
<https://www.thehindu.com/society/history-and-culture/saving-the-precious-geo-heritage-sites-of-visakhapatnam/article33117808.ece>

19. (c)

Explanation:

Statement 1 and 2 are correct:

The force vectors change as the slope angle increases. The gravitational force doesn't change, but the shear force increases while the normal force decreases. **The steepest angle at which rock and soil material is stable and will not move downslope is called the angle of repose**. The angle of repose is measured relative from the horizontal. When a slope is at the angle of repose, the shear force is in equilibrium with the normal force. If the slope becomes just slightly steeper, the shear force exceeds the normal force, and the material starts to move downhill. The shear force exceeds the normal force, and the material starts to move downhill. The angle of repose varies for all material and slopes depending on many factors such as grain size, grain composition, and water content.

Mass-wasting events often have a trigger: something changes that causes a landslide to occur at a specific time. It could be rapid snowmelt, intense rainfall, earthquake shaking, volcanic eruption, storm waves, rapid-stream erosion, or human activities, such as grading a new road. **Increased water content within the slope is the most common mass-wasting trigger**.

X Factor:

Landslides are the third biggest natural disasters in the world, with India experiencing the biggest bulk of them 15 per cent of the country is prone to landslides and India has the highest number of landslide deaths in the world.

Source:

<https://www.indiatoday.in/india/story/indian-researchers-landslide-detection-systems-fatalities-damage-1868763-2021-10-24>

TEST – 9009 (EXPLANATION & SOURCE)

<https://opengeology.org/textbook/10-mass-wasting/>

20. (c)

Explanation:

Both statements are correct:

Bhutan is the only country in the world to officially proclaim the Gross National Happiness (GNH) as the measure of the country's progress. Material progress and technological developments are approached more cautiously taking into consideration the possible harm they might bring to the environment or the other aspects of cultural and spiritual life of the Bhutanese. This simply means material progress cannot come at the cost of happiness. GNH encourages us to think of the spiritual, non-material and qualitative aspects of development.

Source: Human Geography 12th NCERT; Chapter-4; Page- 27

21. (b)

Explanation:

Statement 1 is incorrect: NASA launched CAPSTONE, a microwave oven-sized CubeSat weighing just 55 pounds (25 kg). CAPSTONE, short for Cislunar Autonomous Positioning System Technology Operations and Navigation Experiment, is designed to test a unique, elliptical lunar orbit. **Statement 2 is correct:** CAPSTONE's primary objective is to test and verify the calculated orbital stability of a Near Rectilinear Halo Orbit around the Moon, the same orbit planned for Gateway. NASA's Gateway is a small space station that will orbit around the Moon to provide astronauts with access to the lunar surface. It will feature living quarters for astronauts, a lab for science and research and ports for visiting spacecraft. CAPSTONE is one of the first steps to learn how to operate more robust missions in this unique orbit, thus laying the groundwork for future exploration of our solar system.

Statement 3 is correct: As a pathfinder for Gateway, a Moon-orbiting outpost that is part of NASA's Artemis program, CAPSTONE will help reduce risk for future spacecraft by validating innovative navigation technologies and verifying the

dynamics of this halo-shaped orbit.

X-Factor:

- Chandrayaan 3 is an ISRO (Indian Space Research Organization) mission with the primary objective of putting a lander and rover in the highlands near the south pole of the Moon in 2023. The lander/rover will be similar to the Vikram rover on Chandrayaan 2, with improvements to help ensure a safe landing. It will also carry a communications relay satellite. The rover will carry a seismometer, heat flow experiment, and spectrometers.
- The mission is scheduled to launch in 2023 on a GSLV Mark 3 heavy lift launch vehicle.

Source:https://www.nasa.gov/directorates/spacetech/small_spacecraft/capstone
<https://www.rocketlabusa.com/missions/completed-missions/capstone/>
<https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=CHANDRYN3>

22. (b)

Explanation: Structure of the earth:

Earth's interior can be divided into the crust, upper mantle, lower mantle, outer core, and inner core.

The Crust It is the outermost solid part of the earth. It is brittle in nature. The Mantle The portion of the interior beyond the crust is called the mantle. The mantle extends from Moho's discontinuity to a depth of 2,900 km. The upper portion of the mantle is called asthenosphere. It is considered to be extending upto 400. The core-mantle boundary is located at the depth of 2,900 km. The outer core is in liquid state while the inner core is in solid state.

Pairs 1 and 2 are incorrect: The boundary between the upper crust and the lower crust is termed as 'Conorod boundary'.

The Mohorovicic is the boundary which divides the lower crust and the upper mantle. The density of the mantle is 3.9 g/cm³.

Pairs 3 and 4 are correct: The core has two parts namely the outer core and the inner core. The boundary between the lower mantle and the outer core is called Guttenberg margin. The outer core and inner core are separated by Lehmann boundary.

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X factor:

- The temperature in general increases at the rate of 1°C for every 32 meters towards the earth's interior.
- The earthquake wave velocities helped in understanding the existence of the core of the earth.

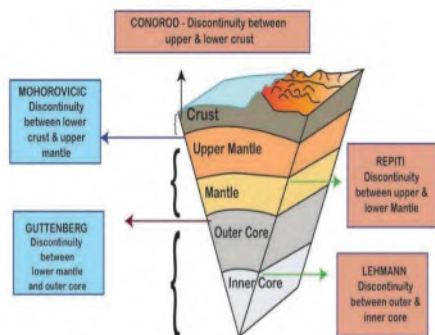


Figure 3.4 Layers of the Earth

Source: SCERT: TN: XI: Geography: Chapter 3: Lithosphere Endogenic process: Page 53, 54.

23. (a)

Explanation:

Statement 1 is correct: Highways are metalled roads connecting distant places. They are constructed in a manner for unobstructed vehicular movement. The **Trans-Continental Stuart Highway connects Darwin (north coast) and Melbourne** via Tennant Creek and Alice Springs in Australia.

Statement 2 is incorrect: The **Northern Atlantic Sea Route** links North-eastern U.S.A. and Northwestern Europe, the two industrially developed regions of the world. The foreign trade over this route is greater than that of the rest of the world combined. One fourth of the world's foreign trade moves on this route. **It is, therefore, the busiest in the world and otherwise, called the Big Trunk Route.**

X-Factor:

- **Big Inch** is a famous pipeline, which carries petroleum from the oil wells of the Gulf of Mexico to the North-eastern States.
- The **Southern Atlantic Sea Route** connects West European and West

African countries with Brazil, Argentina and Uruguay in South America.

Source: NCERT 12th: Fundamentals of Human Geography, Chapter 8, Page no. 68-73

24. (c)

Explanation:

Statement 1 and 2 are correct: The age-sex structure of a population refers to the number of females and males in different age groups. **A population pyramid is used to show the age-sex structure of the population.** The shape of the population pyramid reflects the characteristics of the population. The left side shows the percentage of males while the right side shows the percentage of women in each age group. **The age pyramid shows India's population is young, which, NFHS-5 noted, is typical of developing countries with low life expectancy. The pyramid also shows that fertility has decreased considerably in the last 5.**

Statement 3 is incorrect: India's population remains young, with more than one-fourth aged under 15 years and less than an eighth over 60. There has been only a slight dip in the young population's share in the last five years: **Between the National Family Health Survey-4 (2015-16) and , NFHS-5 (2019-21), the under-15 population has declined by 2 percentage points, from 29% to 27%, while the over-60 population has increased by as many points, from 10% to 12%.**

Over half the population (52%) is below 30, compared to 55.5% in NFHS-4. The NFHS divides the population into 5-year age groups from 0-4 years to 75-79, while those over 80 are counted in a single age group.

Source:

<https://indianexpress.com/article/explained/half-indias-population-under-age-30-nfhs-explained-7910458/>

25. (a)

Explanation:

Option (a) is correct:

Environmental determinism is the belief that the environment, most notably its physical factors such as landforms and

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climate, determines the patterns of human culture and societal development.

Environmental determinists believe that ecological, climatic, and geographical factors alone are responsible for human cultures and individual decisions. Also, social conditions have virtually no impact on cultural development.

The main argument of environmental determinism states that an area's physical characteristics like climate have a substantial impact on the psychological outlook of its inhabitants. These different outlooks then spread throughout a population and help define the overall behavior and culture of a society. For instance, it was said that areas in the tropics were less developed than higher latitudes because the continuously warm weather there made it easier to survive and thus, people living there did not work as hard to ensure their survival.

Another example of environmental determinism would be the theory that island nations have unique cultural traits solely because of their isolation from continental societies.

X-factor:

Possibilism: the theory in geography that human behaviour, and therefore culture, is not merely determined by the environment but by human agency, as a theory it is directly opposed to environmental determinism.

Neo-determinism: A geographer, Griffith Taylor introduced another concept which reflects a middle path (Madhyam Marg) between the two ideas of environmental determinism and possibilism. He termed it as Neodeterminism or stop and go determinism. The concept shows that neither is there a situation of absolute necessity (environmental determinism) nor is there a condition of absolute freedom (possibilism). It means that human beings can conquer nature by obeying it.

Source: Human Geography 12th NCERT; Chapter-1; Page- 4

26. (a)

Explanation:

Statement 1 is correct: Article 1 of General Agreement on Tariffs and Trade (GATT),

1994, requires every WTO member country to accord MFN status (or preferential trade terms with respect to tariffs and trade barriers) to all other member countries.

Statement 2 is incorrect: Accordingly, **India accorded MFN status to all WTO member countries, including Pakistan**, from the date of entry into force of the so-called **Marrakesh Agreement**, establishing the WTO.

During **2019**, after suicide **bombing attack in Pulwama**, the **Indian government has withdrawn "Most Favoured Nation" or MFN status accorded to Pakistan**.

Finance Minister Arun Jaitley said the Ministry of External Affairs will initiate all possible steps to ensure complete isolation from the international community of Pakistan, which is expected to have a hand in the attack carried out by terror outfit Jaish-e-Mohammed.

The primary purpose of the WTO is to open trade for the benefit of all. In that sense, “most favoured” sounds like a contradiction. But even though it suggests special treatment, in the WTO it actually means non-discrimination — that is treating virtually everyone equally. In effect, then, every WTO member is supposed to be “most favoured” for all other WTO members.

X factor:

After becoming a part of WTO, India granted MFN status to Pakistan in 1996, but it was not reciprocated by the latter. After more than a decade, Pakistan offered to extend MFN status to India in 2012 but did not follow through.

Source: NCERT: Fundamentals of Human Geography, Chapter 9, Page 85.

<https://indianexpress.com/article/explained/explained-what-is-mfn-status-how-can-india-hurt-pakistan-by-withdrawing-it-following-pulwama-attack-5585409/>

<https://www.indiatoday.in/india/story/india-withdraws-most-favoured-nation-status-to-pakistan-what-it-means-1456746-2019-02-15>

27. (d)

Explanation:

Option (d) is correct:

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The idea of human development is supported by the concepts of **equity, sustainability, productivity and empowerment.**

Equity refers to making equal access to opportunities available to everybody. The opportunities available to people must be equal irrespective of their gender, race, income and in the Indian case, caste.

Sustainability means continuity in the availability of opportunities. To have sustainable human development, each generation must have the same opportunities.

Productivity here means human labour productivity or productivity in terms of human work. Such productivity must be constantly enriched by building capabilities in people.

Empowerment means to have the power to make choices. Such power comes from increasing freedom and capability. Good governance and people-oriented policies are required to empower people.

X-Factor:

- The Human Development index and the Human Poverty index are two important indices to measure human development used by the UNDP.
- Many of the countries with a high human development score are located in Europe and represent the industrialised western world.

Source: NCERT 12th: Fundamentals of Human Geography, Chapter 4, Page no. 25

28. (c)

Explanation:

Option(c) is correct:

Effects of Volcanic Activities

Destructive effects of volcano:

1. Showers of cinders and bombs can cause damage to life and properties. Sometimes ash can precipitate under the influence of rain and completely cover large areas.
2. The volcanic gases pose potential hazards to people, animals; agriculture, while Sulphur dioxide gas **can lead to acid rain and air pollution.**
3. Massive volcanic eruptions were to blame **for ocean acidification** that wiped out

90 percent of marine biodiversity 250 million years ago.

Positive Effects of Volcanoes:

1. Volcanism **creates new landforms.** Volcanic rocks **yield very fertile soil upon** weathering and decomposition.
2. The Kimberlite rock of South Africa, the **source of diamonds**, is the pipe of an ancient volcano.
3. In the vicinity of active volcanoes, waters in the depth are heated from contact with hot magma **giving rise to springs and geysers.** It will be used for **Geothermal Energy generation.**

X factor: The **Puga valley in Ladakh region and Manikaran (Himachal Pradesh)** are promising spots in India for the **generation of geothermal electricity.**

Source: SCERT: TN: XI: Geography: Chapter 3: Lithosphere Endogenic process: Page 69.

29. (a)

Explanation:

Volcanism, any of various processes and phenomena associated with the surficial discharge of molten rock, pyroclastic fragments, or hot water and steam, including volcanoes, geysers, and fumaroles.

Statement 1 and 3 are correct: Shield Volcanoes: The shield volcanoes are the largest of all the volcanoes on the earth. The Hawaiian volcanoes are the most famous examples. These volcanoes are **mostly made up of basalt**, a type of lava that is **very fluid when erupted**. For this reason, **these volcanoes are not steep.**

Statement 2 is incorrect: Caldera: These are the most explosive of the earth's volcanoes. They are usually so explosive that when they erupt, they tend to collapse on themselves rather than building any tall structure. The collapsed depressions are called calderas. Their explosiveness indicates that the magma chamber supplying the lava is not only huge but is also in close vicinity.

X factor: The Circum-Pacific region:

- Popularly termed the '**Pacific Ring of Fire**', has the greatest concentration of active volcanoes.

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- Volcanic belt and earthquake belt closely overlap along the 'Pacific Ring of Fire'.
- It is estimated to include **two-thirds of the world's volcanoes.**

Source: NCERT: XI: Fundamentals of Physical Geography: Chapter 3: Interior of Earth: Page 26, 27.

30. (d)

Explanation:

Option (d) is correct:

Growth of Population: Change of population in particular area between two points of time is known as growth of population. For example, if we deduct the population of India 2001 (102.70 crore) from population of 2011 (121.02 crore) then we shall get the growth of population (18.15 crores) in actual numbers.

Growth Rate of Population: This is the change of population expressed in percentage.

Natural Growth of Population: This is the population increased by difference between births and deaths in a particular region between two points of time.

Natural Growth = Births – Deaths

Actual Growth of Population: This is Births – Deaths + In Migration – Out Migration

Positive Growth of Population: This happens when the birth rate is more than the death rate between two points of time or when people from other countries migrate permanently to a region.

Negative Growth of Population: If the population decreases between two points of time it is known as negative growth of population. It occurs when the birth rate falls below the death rate or people migrate to other countries.

X-factor:

Crude Death Rate:

- Death rate plays an active role in population change. Population growth occurs not only by increasing birth rate but also due to decreasing death rate.
- Crude Death Rate (CDR) is a simple method of measuring mortality of any area. CDR is expressed in terms of number of deaths in a particular year per thousand of population in a particular region.

Source: Human Geography 12th NCERT; Chapter-2; Page- 10

31. (c)

Explanation:

Most geophysical surveys are aimed at separating out the in order to interpret the geologic structure. It is therefore necessary to make proper allowance for the other factors. **The first two factors imply a variation of gravity with latitude that can be calculated for an assumed shape for Earth.** The third factor, which is the decrease in **gravity with elevation**, due to **increased distance from the center of Earth**, amounts to -0.3086 milligal per meter.

The value of the attraction of gravity or of the potential is determined by the distribution of matter within Earth or some other celestial body. In turn, the **distribution of matter determines the shape of the surface on which the potential is constant.** Measurements of gravity and the potential are thus essential both to geodesy, which is the study of the shape of Earth, and to geophysics, the study of its internal structure

Statement 1 is correct: The major planets, all of **which rotate quite fast**, have large **equatorial bulges**, and their gravity is **dominated by a large increase from equator to pole.** The polar flattening of Jupiter is about 10 percent and was first estimated from telescopic observation by Gian Domenico Cassini about 1664. The results of gravity measurements are crucial to understanding the internal properties of the planets.

Statement 2 is correct: The gravitational force (g) is not the same at different latitudes on the surface. It is greater near the poles and less at the equator. This is because of:

1. the distance from the center at the equator being greater than that at the poles.
2. The gravity values also differ according to the mass of material. The **uneven distribution of mass of material within the earth influences** this value.

X factor:

The reading of the gravity at different places is influenced by many other factors. These

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readings differ from the expected values. Such a difference is called a **gravity anomaly**. Gravity anomalies give us **information about the distribution of mass of the material in the crust of the earth**.

Source: NCERT: XI: Fundamentals of Physical Geography, Chapter 3, Interior of Earth: Page No. 22.

<https://www.britannica.com/science/gravity-physics/Acceleration-around-Earth-the-Moon-and-other-planets#ref210868>

32. (a)

Explanation:

Option (a) is correct:

Orans, or **sacred grove**, are **sacred spaces**, which are **rich in biodiversity** and usually **include a water body**. Communities in **Rajasthan** have been conserving these *Orans* for centuries and their lives have been intricately linked around these spaces. *Orans* are also spaces where herders take their livestock for grazing and are places for communal congregations, festivals, and other social events, the performance of which is linked to agrarian rhythms and the continued commitment of the communities towards environmental conservation.

X-factor:

- **Sacred groves are patches of primeval forest** that some rural communities protect as abodes of deities. Such “ecosystem people” draw their livelihoods from nearby resources and value nature for the ecological services it provides.
- **Colonial resource extraction devastated** the ancient network of sacred groves in India. Thousands of groves survived, however, and some are being newly established because of the ecological benefits that communities derive from them.
- **Empowering ecosystem people to protect** their environment could be key to preserving such pristine spaces. India's Forest Rights Act of 2006, which grants communities rights to manage neighboring forests, is a significant step to this end.

Source:

<https://www.thehindu.com/news/national/other-states/special-story-monsoon-rains-fill-up-traditional-water-harvesting-structures/article65737048.ece>

<http://www.heritageonline.in/natural-heritage-discussion-3/>

<https://www.iccaconsortium.org/index.php/2021/02/27/orans-rajasthan-india/>

<https://www.scientificamerican.com/article/sacred-groves-an-ancient-tradition-of-nature-conservation/>

33. (b)

Explanation:

Statement 1 is incorrect:

The year 2021 was a record one for trade despite the pandemic. In terms of volumes, merchandise trade rose 9.8 per cent, while in dollar terms, it grew 26 per cent. The value of commercial services trade was also up 15 per cent.

India has had a good export run in line with global trends, witnessing record goods exports of \$419 billion, while touching \$250 billion in services exports.

Statement 2 is correct:

In 2021, India's **biggest export item was petroleum and petroleum products — accounting for 15% of total exports** — and their value grew by a whopping 158% (more than three times the overall average rate). And this has happened in a year when crude oil prices have steadily gone up.

Of the 20 commodity groups, only six (highlighted by red circles) had a growth rate higher than the overall average (49%). Within these six, two had insignificant shares. In other words, most of the commodity groups grew at a rate lower than the overall average.

Commodity	2020	2021	2021 over 2020
1. PLANTATION	1,807.71	1,807.00	19.00
2. FISH & FISHED PRODUCTS	25,211.81	26,438.04	28.72
3. MINING PRODUCTS	4,080.20	5,000.00	24.21
4. CEREALS & MISCELLANEOUS	4,080.00	4,000.00	-1.88
5. LEATHER & LEATHER MANUFACTURES	3,000.00	3,000.00	0.00
6. METALS & METALLURGICAL PRODUCTS	18,720.74	20,100.00	14.85
7. CHEMICALS & RELATED PRODUCTS	20,000.00	20,000.00	0.00
8. PLASTIC & RUBBER ARTICLES	4,000.00	4,000.00	0.00
9. ARTICLES OF STONE, PLASTER, CEMENT, ABSORBENT, WICK OR SIMILAR MATERIALS, CERAMIC PRODUCTS, GLASS AND GLASSWARE	4,000.00	4,000.00	0.00
10. TEXTILES & TEXTILE MANUFACTURES	4,000.00	4,000.00	0.00
11. FUEL & MINERAL PRODUCTS	21,000.00	21,000.00	0.00
12. OPTICAL, MEDICAL & SURGICAL INSTRUMENTS	1,770.00	2,000.00	23.73
13. ELECTRONIC EQUIPMENT	4,000.00	4,000.00	0.00
14. TRANSPORT EQUIPMENT	110.00	110.00	0.00
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80. TRANSPORT EQUIPMENT	110.00	110.00	0.00
81. TRANSPORT EQUIPMENT	110.00	110.00	0.00
82. TRANSPORT EQUIPMENT	110.00	110.00	0.00
83. TRANSPORT EQUIPMENT	110.00	110.00	0.00
84. TRANSPORT EQUIPMENT	110.00	110.00	0.00
85. TRANSPORT EQUIPMENT	110.00	110.00	0.00
86. TRANSPORT EQUIPMENT	110.00	110.00	0.00
87. TRANSPORT EQUIPMENT	110.00	110.00	0.00
88. TRANSPORT EQUIPMENT	110.00	110.00	0.00
89. TRANSPORT EQUIPMENT	110.00	110.00	0.00
90. TRANSPORT EQUIPMENT	110.00	110.00	0.00
91. TRANSPORT EQUIPMENT	110.00	110.00	0.00
92. TRANSPORT EQUIPMENT	110.00	110.00	0.00
93. TRANSPORT EQUIPMENT	110.00	110.00	0.00
94. TRANSPORT EQUIPMENT	110.00	110.00	0.00
95. TRANSPORT EQUIPMENT	110.00	110.00	0.00
96. TRANSPORT EQUIPMENT	110.00	110.00	0.00
97. TRANSPORT EQUIPMENT	110.00	110.00	0.00
98. TRANSPORT EQUIPMENT	110.00	110.00	0.00
99. TRANSPORT EQUIPMENT	110.00	110.00	0.00
100. TRANSPORT EQUIPMENT	110.00	110.00	0.00

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X factor:

1. **Balance of trade** records the volume of goods and services imported as well as exported by a country to other countries.
2. If the value of imports is more than the value of a country's exports, the country has a negative **or unfavourable balance of trade**.
3. If the value of exports is more than the value of imports, then the country has a **positive or favourable balance of trade**.

Source: NCERT: Fundamentals of Human Geography, Chapter 9, Page 85.

<https://indianexpress.com/article/opinion/columns/india-must-seize-the-trade-opportunity-opening-7903289/>

<https://indianexpress.com/article/explained/explainspeaking-400-billion-exports-record-india-gdp-growth-7839790/>

34. (a)

Explanation:

Statement 1 is correct: The Lancang-Mekong Cooperation (LMC), a **sub-regional tie-up**, marks the sixth anniversary of the mechanism. The LMC was officially launched at the inaugural LMC Leaders' Meeting in Sanya, South China's Hainan Province, **on March 23, 2016** for cooperation between the **riparian states of the Lancang River and Mekong River**.

- **The Lancang River originates on the Qinghai-Tibet Plateau in southwestern China.** It is called the Mekong River as it flows through the other five countries before emptying into the sea. The purpose of the grouping is for China to manage water flow from its hydropower dams with the other riparian states. **As a platform for discussion and sharing ideas**, it aims to push forward pragmatic projects to benefit the six countries.
- **Statements 2 and 3 are incorrect: Its members include the six countries along the Lancang-Mekong River: China, Thailand, Cambodia, Laos, Myanmar and Vietnam.**

X-Factor:

- The Mekong River is the longest river in Southeast Asia. The river has a length of approximately 4,900 km, flowing from its source on the Tibetan Plateau in China

through Myanmar, Lao PDR, Thailand, Cambodia and Viet Nam via a large delta into the sea.

Source:

<https://www.globaltimes.cn/page/202203/1256659.shtml>

<https://www.thehindu.com/news/international/chinas-top-diplomat-visits-myanmar-amid-opposition-protest/article65593101.ece>

<https://indianexpress.com/article/world/chinas-top-diplomat-visits-myanmar-amid-opposition-protest-8005582/>

<https://www.mrcmekong.org/about/mekong-basin/geography/>

35. (b)

Explanation:

Pair 1 is correctly matched and pair 4 is incorrectly matched: Tripoli and Abadan are oil ports. Oil Ports deal in the processing and shipping of oil. Some of these are tanker ports and some are refinery ports. Maracaibo in Venezuela, Esskhira in Tunisia, and **Tripoli in Lebanon** are tanker ports. **Abadan on the Gulf of Persia (in Iran)** is a refinery port.

Pair 3 is correctly matched: Memphis is an inland port. Inland Ports are located away from the sea coast. They are linked to the sea through a river or a canal. For example, **Manchester is linked with a canal; Memphis is located on the river Mississippi;** Rhine has several ports like Mannheim and Duisburg; and Kolkata is located on the river Hoogli, a branch of the river Ganga.

Pair 2 is incorrectly matched: Calais is a ferry port (packet station) in France. These packet stations are exclusively concerned with the transportation of passengers and mail across water bodies covering short distances. These stations occur in pairs located in such a way that they face each other across the water body.

X-Factor:

Important Ports of the World:

- Havana - Cuba
- Jebel Ali- Dubai, United Arab Emirates
- Rotterdam- The Netherlands
- Port Klang- Malaysia
- Antwerp - Belgium

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- Hanshin Port- Japan
- Piraeus, Greece
- Algeciras, Spain

Source: NCERT 12th: Fundamentals of Human Geography, Chapter 9, Page no. 89

36. (a)

Explanation:

statements 1 is correct: The factors which affect the distribution of temperature of ocean water are:

(i) Latitude: the **temperature of surface water decreases from the equator towards the poles** because the amount of insolation decreases poleward.

Statements 2 and 3 are incorrect: The factors which affect the distribution of temperature of ocean water are:

(ii) Unequal distribution of land and water: the **oceans in the northern hemisphere receive more heat due to their contact with larger extent of land** than the oceans in the southern hemisphere.

The oceans in the northern hemisphere record relatively higher temperature than in the southern hemisphere. The **highest temperature is not recorded at the equator but slightly towards north of it**. The average annual temperatures for the northern and southern hemisphere are around 19° C and 16° C respectively. This variation is due to the unequal distribution of land and water in the northern and southern hemispheres.

X factor: The enclosed seas in the low latitudes record relatively higher temperature than the open seas; whereas the enclosed seas in the high latitudes have lower temperature than the open seas.

Source: NCERT: Fundamentals of Physical Geography: Chapter 13: Page:111, 112.

37. (d)

Explanation:

Option (d) is correct:

Mica is a mineral made up of a series of plates or leaves. **It splits easily into thin sheets.**

These sheets can be so thin that a thousand can be layered into a mica sheet of a few

Centimetres high. Mica can be clear, black, green, red yellow or brown. Due to its excellent

Di-electric strength, **low power loss factor, insulating properties** and resistance to high

Voltage, mica is one of the most indispensable minerals used in electric and electronic

Industries. **Andhra Pradesh is India's largest mica producer.** Mica deposits are found in the northern edge of the Chota Nagpur plateau. Koderma Gaya – Hazaribagh belt of Jharkhand is the leading producer.

X-factor:

Steel Plants	Year of Establishment	Collaborating Country	Location
ISCO	1870-1913 (Private Initially)	(Nationalised in 1972)	Kulti and Burnpur (W.B.)
TISCO	1907 (Private ownership)	Jamshedji Tata	Jamshedpur (Jharkhand)
VISL	1923 Private as MISCO	Nationalised later	Bhadrahati (Karnataka)
Bhilai	1959 (Public Sector)	Soviet Union	Bhilai (M.P.)
Bokaro	1972 (Public Sector)	Soviet Union	Bokaro (Jharkhand)
Durgapur	1959 (Public Sector)	U.K.	Durgapur (W.B.)
Rourkela	— (Public Sector)	Germany	Rourkela (Odisha)
Vishakhapatnam	— (Public Sector)	—	Vishakhapatnam Andhra Pradesh
Salem	— (Public Sector)	—	Salem (T. Nadu)
Vijay Nagar	— Public Sector	—	Karnataka

Source: NCERT, Geography Class 10, Chapter5, page no 6 Chapter 6-page no. 94

38. (b)

Explanation:

Enhanced Rock Weathering is a proposed **Carbon Dioxide Removal technology** involving the application of crushed silicate rocks, such as basalt, to agricultural soils with potential co-benefits for crops and soils, and mitigation of ocean acidification.

Statement 1 is incorrect: Natural rock weathering absorbs around 0.3% of global fossil fuel emissions, **enhanced weathering can provide** a boost to remove even more CO₂ from our atmosphere. As enhanced **weathering makes water more alkaline**, it can help **counteract ocean acidification**. And adding minerals to soils can boost nutrient levels, improving crop yields and helping restore degraded agricultural soils

Statement 2 is correct: Indonesia ranks second in fossil fuel emissions per unit

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of energy behind India, ranks low in human health impacts, because **its tropical climate and perennial crops favour higher weathering rates, meaning that less grinding is required to achieve an equivalent amount of Carbon dioxide removal.**

X Factor:

Arable farms already apply crushed rock in the form of limestone to reverse acidification of soils caused by farming practices, such as the use of fertilisers. And there is a long history of small-scale farming using silicate rocks to improve crop yields in highly-weathered soils in Africa, Brazil and Malaysia.

Swapping silicate for limestone, and increasing the application rate, would do the same job to help tackle acidification, but help capture CO₂ from the atmosphere at the same time.

Source:

<https://www.nature.com/articles/s43247-022-00436-3>

<https://www.carbonbrief.org/guest-post-how-enhanced-weathering-could-slow-climate-change-and-boost-crop-yields/#:~:text=As%20natural%20rock%20weathering%20absorbs,can%20help%20counteract%20ocean%20acidification.>

39. (d)

Explanation:

Statement 1 is correct:

With a long coastline of 7,516.6 km, India is dotted with 12 major and 200 notified nonmajors (minor/intermediate) ports. These major ports handle 95 per cent of India's foreign trade.

Kandla in Kutch was the **first port developed soon after Independence to ease the volume of trade on the Mumbai port, in the wake of loss of Karachi port to Pakistan after the Partition.** Kandla, also known as the **Deendayal Port**, is a **tidal port**. It caters to the convenient handling of exports and imports of highly productive granary and industrial belt stretching across UT of Jammu and Kashmir, and the states of Himachal Pradesh, Punjab, Haryana, Rajasthan and Gujarat.

Statement 2 is correct: Kolkata is an **inland riverine port**. This port serves a very large and rich hinterland of Ganga Brahmaputra basin. Being a tidal port, it requires constant dredging of Hooghly.

Prime Minister Narendra Modi on Sunday (January 12) **renamed** the **Kolkata Port Trust** after Bharatiya Jana Sangh founder **Dr Syama Prasad Mookerjee**, at an event to mark its **150th anniversary**.

The **Kolkata port is the only riverine port in the country**, situated 203 km from the sea. The river Hooghly, on which it is located, has many sharp bends, and is considered a difficult navigational channel. Throughout the year, dredging activities have to be carried out to keep the channel open.

Statement 3 is correct: The Indian Government plays an important role in supporting the ports sector. It has allowed **Foreign Direct Investment (FDI) of up to 100% under the automatic route for port and harbour construction and maintenance projects**. It has also facilitated a 10-year tax holiday to enterprises that develop, maintain and operate ports, inland waterways and inland ports.

X factor:

1. India has **12 major and 205 notified minor and intermediate ports**. Under the National Perspective Plan for Sagarmala, six new mega ports will be developed in the country.
2. India is the **sixteenth-largest maritime country in the world** with a coastline of about 7,517 kms.

Source: NCERT: Contemporary India II: Chapter 7: Page: 85, 86.

<https://indianexpress.com/article/explained/significance-of-the-kolkata-port-renamed-by-pm-modi-6213004/>

<https://www.ibef.org/industry/ports-india-shipping>

40. (b)

Explanation:

The art of crafting is a 400-year-old tradition. The artisans who make the toys are referred as Aryakhastriyas (also known as Nakarshalu), who have their mention in the Brahmanda Purana. These craftsmen are said to have migrated from Rajasthan in

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the 16th century to Kondappali and claims their origin to Muktharishi, a sage endowed with skills in arts and crafts by Lord Shiva.

Statements 1 and 2 are incorrect: The Kondapalli toys are **made from soft wood known as Tella Poniki** which are found in nearby Kondapalli Hills. Each part is carved out separately. Then makku- a paste of tamarind seed powder & sawdust is used to join pieces together, add details and finish the toys, in Kondapalli of Krishna district, in the state of **Andhra Pradesh**. These toys were one of the varieties of toys assembled in the houses during the festivals of Sankranti and Navratri and is referred as Bommala Kolu.

Statement 3 is correct: Kondapalli toys are famous for their light weight, vibrant colors and age-old production techniques. Themed around mythology, rural life and animals, these toys exhibit joyous and realistic expressions. **A swift influence of Islamic and Rajasthani forms can be seen in the woodwork and detailing of each toy.**

Statement 4 is correct: This craft was registered as one of the Geographical Indication handicraft from Andhra Pradesh as per Geographical Indications of Good (Registration and Protection) Act, 1999.

X-Factor:

- **The indigenous toys from each region of India:**
- **Thanjavur doll:** The craft of the doll is an ancient skill where few or no tools are used to handcraft the figurines. The unique feature about the Thanjavur Doll is that it is a combination of roly-poly and bobblehead dolls.
- **The wooden toys of Varanasi:** Uttar Pradesh's Varanasi and Mirzapur are renowned for their wooden toys which are made of the wood. Made without any joints, the beautiful toys are an ancient craft and are cast in shapes of animals, birds, musicians and dancers.
- **Asharikandi Teracotta toys:** The traditional toys in Assam are made with clay, bamboo, pith, cloth and mud in the figures of brides and grooms, animals, birds and deities. But one of the most famous dolls found in the region is from Asharikandi.

- **Natungram doll of West Bengal:** Crafted from a piece of wood the doll gets the face and its attire painted on it and preserves a rustic beauty. **The Gour-Nitai dolls** are one of the distinct types of these dolls which came from the wave of Bhakti movement in Bengal.
- **Dolls of Vilachery of Tamil Nadu:** Made up of clay and paper mache, **golu dolls** for the Navratri season. Families line up their homes with **golu** dolls of various shapes and sizes. The skilled craftsmen mould the dolls as gods, goddesses, and characters from epics, folk tales and mythology. They are then sun-dried. The artisans then paint them with two coats of bright colours and can complete a doll in three to five days.

Source:

<https://cultureandheritage.org/2022/05/kondapalli-wooden-toys-of-andhra.html>
<https://www.thebetterindia.com/236824/indian-desi-native-toys-old-heritage-buy-karnataka-channapatna-toys-geographical-indication-handicraft-dolls-sai87/>
<https://www.thehindu.com/life-and-style/traditional-telugu-foods-are-a-click-away/article65774590.ece>

41. (d)

Explanation:

All statements are correct:

In the **Union Budget 2021-22**, presented on 1st February 2021, the finance minister announced an outlay of **INR 1.97 Lakh Crores** for the **Production-Linked Incentive (PLI) Schemes for 14 key sectors, to create national manufacturing champions and generate employment opportunities for the country's youth**. This means that minimum production in India as a result of PLI Schemes is expected to be over US\$ 500 billion in 5 years.

PLI Schemes are a **cornerstone of the Government's push for achieving an Atmanirbhar Bharat**. The objective is to **make domestic manufacturing globally competitive and to create global Champions in manufacturing**. The strategy behind scheme is to offer **companies incentives on incremental sales from products manufactured in India**, over the base year. They have been specifically

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designed to **boost domestic manufacturing in sunrise and strategic sectors**, curb cheaper imports and **reduce import bills**, improve **cost competitiveness** of domestically manufactured goods, and enhance domestic capacity and exports.

X-factor:

In addition to the three schemes announced earlier in March 2020, GoI has further introduced the following 10 new PLI schemes in November 2020:

March 2020

- Key Starting Materials (KSMs)/Drug Intermediates (DIs) and Active Pharmaceutical Ingredients (APIs): Department of Pharmaceuticals
- Large Scale Electronics Manufacturing: Ministry of Electronics and Information Technology
- Manufacturing of Medical Devices: Department of Pharmaceuticals

November 2020:

- Electronic/Technology Products: Ministry of Electronics and Information Technology
- Pharmaceuticals drugs: Department of Pharmaceuticals
- Telecom & Networking Products: Department of Telecommunications
- Food Products: Ministry of Food Processing Industries
- White Goods (ACs & LED): Department for Promotion of Industry and Internal Trade
- High-Efficiency Solar PV Modules: Ministry of New and Renewable Energy
- Automobiles & Auto Components: Department of Heavy Industry
- Advance Chemistry Cell (ACC) Battery: Department of Heavy Industry
- Textile Products: MMF segment and technical textiles: Ministry of Textiles
- Specialty Steel: Ministry of Steel

September 2021

- Drones and Drone Components: Ministry of Civil Aviation

Source:

<https://www.pib.gov.in/PressReleasePage.aspx?PRID=1710134>

<https://www.investindia.gov.in/production-linked-incentives-schemes-india>

42. (a)

Explanation:

Option (a) is correct:

Volcanic intrusive landforms:

The lava that is released during volcanic eruptions on cooling develops into igneous rocks. Depending on the location of the cooling of the lava, igneous rocks are classified as volcanic rocks (cooling at the surface) and plutonic rocks (cooling in the crust). The lava that cools within the crustal portions assumes different forms. These forms are called intrusive forms.

Laccoliths: These are **large dome-shaped intrusive bodies with a level base and connected by a pipe-like conduit from below**. It resembles the surface **volcanic domes of composite volcano, only these are located at deeper depths**. It can be regarded as **the localized source of lava that finds its way to the surface**. The Karnataka plateau is spotted with domal hills of granite rocks. Most of these, now exfoliated, are examples of laccoliths or batholiths.

Batholiths: A large body of magmatic material that cools in the deeper depth of the crust develops in the form of large domes. They appear on the surface only after the denudational processes remove the overlying materials. They cover large areas, and at times, assume depth that may be several km. These are granitic bodies. Batholiths are the cooled portion of magma chambers.

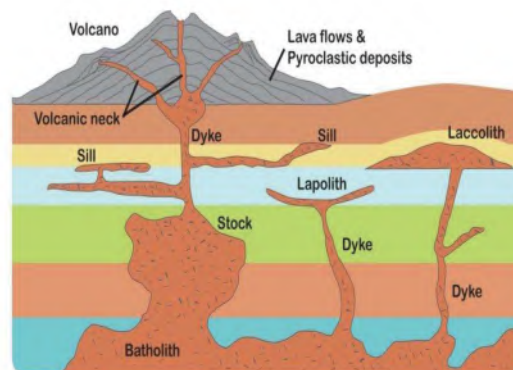


Figure 3.26 Intrusive Volcanic structure

X factor:

www.ksgindia.com

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1. The lava that is released during volcanic eruptions on cooling develops into igneous rocks. The cooling may take place either on reaching the surface or also while the lava is still in the crustal portion.
2. Depending on the location of the cooling of the lava, **igneous rocks** are classified as **volcanic rocks (cooling at the surface)** and **plutonic rocks (cooling in the crust)**.

Source: NCERT: XI: Fundamentals of Physical Geography: Chapter 3: Page 27.

43. (d)

Explanation:

Statement 1 is correct: Leprosy is a chronic infectious disease caused by a **bacteria bacillus, Mycobacterium leprae**. M leprae multiplies slowly and the incubation period of the disease, on average, is 5 years. Symptoms may occur within 1 year but can also take as long as 20 years or even more.

Statement 2 is correct: Leprosy mainly affects the skin, the peripheral nerves, mucosa of the upper respiratory tract, and the eyes. **Some affected individuals have damage to internal organs and tissues.**

Statement 3 is correct: Leprosy is likely **transmitted via droplets, from the nose and mouth, during close and frequent contact with untreated cases.** Untreated, leprosy can cause progressive and permanent damage to the skin, nerves, limbs, and eyes.

Statement 4 is correct: India achieved the **goal set by the National Health Policy, 2002 of elimination of leprosy as a public health problem, defined as less than 1 case per 10,000 population, at the National level in December 2005.** But many new cases started getting reported post 2005. India has the highest number of new leprosy cases in the world, followed by Brazil and Indonesia. Every year, over 200,000 such cases are detected globally and India accounts for more than half of these, according to the World Health Organization (WHO).

X-Factor:

- Leprosy is endemic in several states and union territories of India, with the annual

case detection rate of 4.56 per 10 000 population.

- The **National Leprosy Eradication Programme (NLEP)** is a Centrally Sponsored Health Scheme has been implemented with the major objective of reducing the disease burden, prevention of disability and to improve awareness among the mass about Leprosy and its curability. Multi Drug Therapy (MDT) is an important instrument to reduce the burden of active cases of Leprosy.
- India continues to account for 60% of new cases reported globally each year and is among the 22 “global priority countries” that contribute 95% of world numbers of leprosy warranting a sustained effort to bring the numbers down.
- NLEP annual reports of the last 4 years have consistently observed that the four states/UTs (Orissa, Chandigarh, Delhi, and Lakshadweep), which achieved elimination earlier in 2011–2012, have shown a prevalence of >1 per 10,000 population.

Source:<https://www.downtoearth.org.in/bl-og/health/why-leprosy-has-resurfaced-in-india-63403><https://www.who.int/news-room/fact-sheets/detail/leprosy>
<https://www.ncbi.nlm.nih.gov/pmc/article/s/PMC5885632/>

44. (c)

Explanation:

Features associated with Convergent plate boundary:

1. Convergent plate boundary is the margin where two plates collide with one another.
2. **Ocean trenches** are associated with the Convergent plate boundary.
3. A **trench** is a narrow and deep depression of the ocean floor. It is formed when the oceanic plate slides down underneath continental plate as the oceanic plate is denser than the continental plate.

Features associated with Divergent Plate Boundary:

1. Divergent plate boundary is the margin where two plates move apart.
2. **Narrow oceans** represent young divergent boundaries.

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3. A **submarine mountain ridge** is formed through the fissures in the oceanic crust when the plates move apart.
4. A **rift valley** is formed when two plates move apart. If a divergent boundary runs through the continent, the continent splits apart and a rift valley is formed.

X factor:

The **Mid-Atlantic Ridge** is an ideal example of a submarine mountain ridge in the Atlantic Ocean. It is the **longest mountain ridge in the world**.

Source: SCERT: TN: XI: Geography: Chapter 3: Page No. 58, 59.

45. (b)

Explanation:

Option (b) is correct:

Sedimentary rocks:

The word 'sedimentary' is derived from the Latin word sedimentum, which means settling.

Rocks (igneous, sedimentary and metamorphic) of the earth's surface are exposed to denudational agents, and are broken up into various sizes of fragments. Such fragments are transported by different exogenous agencies and deposited. These deposits through compaction turn into rocks. This process is called lithification.

Depending upon the mode of formation, sedimentary rocks are classified into three major groups:

- (i) **Mechanically formed** — sandstone, conglomerate, limestone, shale, loess etc. are examples;
- (ii) **Organically formed**— geyserite, chalk, limestone, coal etc. are some examples;
- (iii) **Chemically formed** — chert, limestone, halite, potash etc. are some examples.

X factor:

Table 3.2 Metamorphosis processes

Igneous/ Sedimentary rock	Influence	Metamorphosed rock
Granite	Pressure	Gneiss
Clay, Shale	Pressure	Schist
Sandstone	Heat	Quartzite
Clay, Shale	Heat	Slate or Phyllite
Coal	Heat	Anthracite or Graphite
Coal	Heat and Pressure	Diamond
Limestone	Heat	Marble

Source: NCERT: XI: Fundamentals of Physical Geography: Chapter 5: Page 42.,

TN SCERT: XI: Geography: page 72.

46. (c)

Explanation:

Option (c) is correct: All processes that move, elevate or build up portions of the earth's crust come under diastrophism.

: Diastrophism include following processes:

- (i) **Orogenic processes** involving mountain building through severe folding and affecting long and narrow belts of the earth's crust;
- (ii) **Epeirogenic processes** involving uplift or warping of large parts of the earth's crust;
- (iii) **Earthquakes** involving local relatively minor movements;
- (iv) **Plate tectonics** involving horizontal movements of crustal plates.

X factor:

1. **Orogeny** is a **mountain building process** whereas **epeirogeny is continental building process**.
2. Through the processes of orogeny, epeirogeny, earthquakes and plate tectonics, there can be faulting and fracturing of the crust.

Source: NCERT: XI: Fundamentals of Physical Geography: Chapter 6: Page 46.

47. (c)

Explanation:

Option (c) is correct:

The ISRO has initiated '**Project NETRA**' to **monitor space debris**. The domestic surveillance system would provide firsthand information on the status of debris, which would aid further planning on protecting space assets.

X-Factor:

- Space debris encompasses both natural meteoroid and artificial (human-made) orbital debris. Meteoroids are in orbit about the sun, while most artificial debris is in orbit about the Earth (hence the term "orbital" debris).
- Orbital debris is any human-made object in orbit about the Earth that no longer serves a useful function. Such debris includes nonfunctional spacecraft, abandoned launch vehicle stages,

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mission-related debris, and fragmentation debris.

- ISRO's System for Safe and Sustainable Space Operations & Management (IS4OM) would better help with space traffic monitoring.

Source:

<https://www.thehindu.com/news/national/andhra-pradesh/reducing-debris-key-to-sustainability-in-space-ap-governor/article65971845.ece>

https://www.nasa.gov/mission_pages/station/news/orbital_debris.html

<https://www.newindianexpress.com/states/karnataka/2022/jul/12/isro-launches-worlds-first-facility-to-track-space-debris-safeguard-assets-2475563.html>

48. (b)

Explanation:

Spring tides:

The position of both the sun and the moon in relation to the earth has direct bearing on tide height.

Statement 1 is incorrect and Statement 2 is incorrect: When the sun, the moon and the earth are in a straight line, the height of the tide will be higher. These are called spring tides and they occur twice a month, one during the full moon period and another during the new moon period.

Neap tides: Normally, there is a seven-day interval between the spring tides and neap tides. At this time the sun and moon are at right angles to each other and the forces of the sun and moon tend to counteract one another. The Moon's attraction, though more than twice as strong as the sun's, is diminished by the counteracting force of the sun's gravitational pull.

Statement 3 is correct: Tides of Bay of Fundy, Canada the highest tides in the world occur in the Bay of Fundy in Nova Scotia, Canada. The tidal bulge is 15 - 16 m. Because there are two high tides and two low tides every day (roughly a 24-hour period); a tide must come in within about a six-hour period.

Source: NCERT: Fundamentals of Physical Geography: Chapter 14: Page: No. 118.

49. (c)

Explanation:

Option(c) is correct:

Ocean is dominant in the southern hemisphere between the latitudes 40°S and 60°S. Hence the westerlies are so powerful and persistent that the sailors used such expressions as "Roaring Forties", "Furious Fifties" and "Screaming or Shrieking Sixties" for these high velocity winds in the latitudes of 40°, 50° and 60° respectively.

Westerlies flow towards the pole from sub tropics and turn towards right and left in the northern hemisphere and southern hemisphere respectively.

X factor:

There are two important bases on which the pressure belts are formed.

1. Temperature: pressure belts are formed due to high and low temperature respectively.

2. Dynamism: pressure belts are formed due to movement and collision of wind systems.

Source: SCERT: TN: XI: Geography: Chapter: 6: Atmosphere: Page: 151-153.

50. (d)

Explanation:

Statements 1 is incorrect: El Nino is a phenomenon that occurs in the equatorial Pacific Ocean characterized by a positive sea surface temperature departure from normal (1971-2000 base period) in the region lying within the latitude 5°N to 5°S and longitudes 120° W to 170°W . This phenomenon occurs every two to seven years.

Pacific Ocean in Normal situation: Near equator the water of the Pacific Ocean is warmer in the western side and cooler in the eastern side due to upwelling of the cold current.

Pacific Ocean in el nino situation: Near equator the warm water in the Pacific Ocean extends from western side to eastern side suppressing the upwelling of the cold water.

Air (Walker) circulation is dominant in the eastern part of the Pacific Ocean. The air ascends in the warm eastern Pacific Ocean.

Statements 2 and 3 are incorrect: During El nino South East Asia experiences severe drought and forest fire. Peru in

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South America receives heavy rainfall during El Nino, i.e. The **West coast of South America receives heavy rainfall. Australia experiences dry weather conditions.**

X factor:

Peruvian fishermen named the weather phenomenon El Nino meaning 'little boy' or 'New born Christ' and La Nina meaning 'Little girl' as the phenomenon was first noticed during Christmas time.

Source: SCERT: TN: XI: Geography: Chapter: 5: Hydrosphere: Page:130-133.

51. (c)

Explanation:

Option(c) is correct:

Heating of the atmosphere is an indirect process.

The processes are:

a. Terrestrial radiation:

The solar radiation reflected by the earth's surface is called 'Terrestrial radiation'. Terrestrial radiation supplies more heat energy to the atmosphere due to its long wavelength.

b. Conduction

The **heat energy from the earth's surface is transferred to the lower atmosphere which is directly in contact with the surface by the process of conduction.**

c. Convection and advection

The movement of air molecules in vertical and horizontal direction is called 'convection and advection' respectively. This movement carries heat energy to the various parts of the earth and at different altitudes.

X factor:

The **surface of the earth is heated by the sun's rays in the form of short-wave radiation. The heat received by the earth is called 'Solar Radiation' or 'Insolation'.**

Source: SCERT: TN: XI: Geography: Chapter:6: Atmosphere: Page:143.

52. (d)

Explanation:

Statement 1 is incorrect: Periyar river has a length of 244 km and is the longest river in the Kerala. It is a perennial river and used for supplying drinking water. It is popularly called as the lifeline of Kerala. **The river**

originates in the Shivgiri hills of Western Ghats and flow through the Periyar National Park and reaches the Periyar Lake. The water from here then flows into the Vembanad Lake and then finally to the Arabian Sea.

Statement 2 is incorrect: The river is surrounded by the **Periyar Tiger Reserve & National Park.**

Statement 3 is incorrect: The main tributaries of Periyar are Muthirapuzha, Mullayar, Cheruthoni, Perinjankutti and the Edamala rivers. **The largest hydro electricity project called as Idukki dam is commissioned here which is biggest in Asia.** There are some other dams on this river which includes Neriyaamangalam, Mullaperiyar, Pallivasal etc.

X-Factor:

There are many important places situated on the bank of the Periyar river such as the birth place of Sankaracharya at Kaladi, Thattekad Bird Sanctuary etc.

Located at a distance of 15 km from Tiruchirapalli, the **Kallanai Dam** is built across the flowing **River Kaveri.**

Source:

<https://indianexpress.com/article/cities/thiruvananthapuram/tamilnadu-to-open-three-shutters-of-mullaperiyar-dam-today-idukki-on-alert-8072115/>

<https://www.britannica.com/place/Periyar-River>

<https://kerala.me/environment/lakes-and-rivers/periyar>

53. (a)

Explanation:

Statements 1 is correct: The **troposphere** is the lowermost layer of the atmosphere. It **extends approximately to a height of 8 km from the poles and 18 km from the equator.** The height of the troposphere changes seasonally also. It increases during summer and decreases during winter.

Statement 2 is incorrect: The Stratosphere is the second layer of the atmosphere found above the troposphere. It approximately extends up to a height of 50 km from the earth's surface. Temperature is constant up to a height of 20 km and increases gradually up to the stratopause where temperature is nearly -4°C.

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The **lower part of this layer is highly concentrated with ozone gas** which is called '**ozonosphere**'. It prevents the ultra-violet rays from the Sun from entering into the lower part of the atmosphere as the rays are highly harmful; it causes skin cancer and other ill effects to living organisms. But the ozone layer safeguards life on the earth.

Statement 3 is incorrect: The **mesosphere** is the third layer of the atmosphere found approximately up to a height of 85 km above the surface of the earth. It is the **coldest layer of the atmosphere**. The temperature decreases with increase of altitude due to the absence of ozone.

X factor:

1. Ozone (O₃) is a form of oxygen that combines three atoms into each molecule.
2. It absorbs and filters the harmful ultraviolet B radiation coming from the sun.
3. This way the ozone layer protects all life on earth. However, ozone is harmful when it develops near the ground. It causes health problems like asthma and other respiratory illnesses.

Source:

SCERT: TN: XI: Geography: Chapter: 6: Atmosphere: Page:139, 140.

54. (c)

Explanation:

Option(c) is correct:

Local wind systems influence the weather pattern wherever they blow.

Chinook: Warm dry westerly off the Rocky Mountains.

Karaburan: 'Black storm' a spring and summer katabatic wind of **central Asia**.

Mistral: Cold northerly from central France and **the Alps to Mediterranean**.

Sirocco, warm, humid wind occurring over the northern **Mediterranean Sea** and southern Europe, where it blows from the south or southeast and brings uncomfortably humid air. The sirocco is produced on the east sides of low-pressure centers that travel eastward over the southern Mediterranean. It originates over North Africa as a dry wind and picks up moisture as it crosses the Mediterranean.



Figure 6.18 Local winds of the world

X factor:

Loo: Hot and dry wind which blows over plains of India and Pakistan.

Harmattan: Dry northerly wind across central Africa.

Nor'easter: Strong winds from the northeast in the eastern United States, especially New England.

Source: SCERT: TN: XI: Geography: Chapter: 6: Atmosphere: Page:158-159.<https://www.britannica.com/science/sirocco-wind>

55. (c)

Explanation:

Option (c) is correct:

Coriolis Force:

The rotation of the earth about its axis affects the direction of the wind. This force is called the Coriolis force after the French physicist who described it in 1844.

It deflects the wind to the right direction in the northern hemisphere and to the left in the southern hemisphere. The deflection is more when the wind velocity is high.

The Coriolis force is directly proportional to the angle of latitude. It is maximum at the poles and is absent at the equator. The Coriolis force acts perpendicular to the pressure gradient force.

The pressure gradient force is perpendicular to an isobar. The higher the pressure gradient force, the more is the velocity of the wind and the larger is the deflection in the direction of wind. As a result of these two forces operating perpendicular to each other, in the low-pressure areas the wind blows around it.

At the equator, the Coriolis force is zero and the wind blows perpendicular to the

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isobars. The low pressure gets filled instead of getting intensified. That is the reason why tropical cyclones are not formed near the equator.

X factor:

Frictional Force: It affects the speed of the wind. It is greatest at the surface and its influence generally extends up to an elevation of 1 - 3 km. Over the sea surface the friction is minimal.

Source: NCERT: XI: Fundamentals of Physical Geography: Page:86,87.

56. (b)

Explanation:

Option (b) is correct:

The theory of plate tectonics proposes that the earth's lithosphere is divided into seven major and some minor plates. Young Fold Mountain ridges, trenches, and/or faults surround these major plates. Some important minor plates are listed below:

- (i) **Cocos plate: Between Central America and Pacific plate**
- (ii) **Nazca plate: Between South America and Pacific plate**
- (iii) **Arabian plate: Mostly the Saudi Arabian landmass**
- (iv) **Philippine plate: Between the Asiatic and Pacific plate**
- (v) **Caroline plate: Between the Philippine and Indian plate (North of New Guinea)**
- (vi) **Fuji plate: North-east of Australia.**

These plates have been constantly moving over the globe throughout the history of the earth.

TRIBE PLACE

Santhal	Jharkhand and West Bengal
Toto	West Bengal
Bodo	Assam
Bhils	Rajasthan
Gonds	Madhya Pradesh

X-Factor:

Major plates:

- Antarctica and the surrounding oceanic plate
- North American (with western Atlantic floor separated from the South American plate along the Caribbean islands) plate
- South American (with western Atlantic floor separated from the North American plate along the Caribbean islands) plate
- Pacific plate
- India-Australia-New Zealand plate
- Africa with the eastern Atlantic floor plate

Source: NCERT Class 11th Fundamentals of physical geography, Chapter 4, Page no. 35-36

57. (b)

Explanation:

Option (b) is correct

Pair 1 is correct Lepcha: Lepcha is a tribe of Himalayan range live at the North-East corner of India. They largely reside at Meghalaya, Arunachal Pradesh, Bhutan, Sikkim and Darjeeling. They have also migrated in other North-Eastern States for economic reason.

Pair 2 is incorrect: Bhotia Tribe is one of the predominating tribes of Uttarakhand. They reside in the Almora, Chamoli, Pithoragarh and Uttarkashi districts of Uttarakhand. They are also found in Sikkim area.

Pair 3 is correct Mishmi: The Mishmi or Deng people of Tibet and Arunachal Pradesh are an ethnic group comprising mainly three tribes: Idu Mishmi (Idu Lhoba); Digaro tribe (Taraon, Darang Deng), and Miju Mishmi (Kaman Deng). The Mishmis occupy the northeastern tip of the central Arunachal Pradesh in Upper and Lower Dibang Valley, Lohit and Anjaw Districts.

Pair 4 is incorrect Sumi: The Sumi Naga also known as Sema Naga are a major Naga ethnic group in the Indian state of Nagaland. The Sumis mainly inhabit Zunheboto District, parts of Niuland District and Kiphire District although many have spread and are now living in a few more districts within Nagaland.

Some other important tribes of India:

X factor :

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- Odisha has come up with an encyclopedia on tribes documenting their age-old and unique traditions before they get entirely vanished from the circulation.
- Tribes, as per 2011 census, account for 22.85% of the Odisha's total population. Odisha is home to third largest tribal population of India, but it is the most diverse indigenous communities found in the country.
- The State has 62 tribes including 13 particularly vulnerable tribal groups.
- Five edited volumes of 'Encyclopedia of Tribes in Odisha' published by Scheduled Castes and Scheduled Tribes Research and Training Institute and Odisha State Tribal Museum in the Museum complex was released by Chief Minister of Odisha.

Source: Book: NCERT Class 11, India Physical Environment Geography Chapter 2, Structure and Physiography

<https://www.thehindu.com/news/national/other-states/odisha-comes-up-with-encyclopedia-on-tribes/article65937550.ece>

58. (a)

Explanation:

Option(a) is correct:

Indian Virtual Herbarium is a **database of dried plants that maximizes the usefulness of the collections**. Apart from digital images of **the herbarium specimens**, the label data on each species include all information about the herbarium specimen such as family, genus, species, author citation, sub-species, variety (if any), collector, collection number, collection date, herbarium region, localities, plant description, habitat and comments (if any) are available.

Source:

<https://bsi.gov.in/page/en/Virtual-Herbarium>

<https://www.thehindu.com/news/national/indian-virtual-herbarium-biggest-database-of-countrys-flora-grabbing-eyeballs-across-the-world/article65737063.ece>

59. (c)

Explanation:

Stony coral tissue loss disease: Stony Coral Tissue Loss Disease is a rapidly spreading disease affecting over 20 species of hard corals in the Caribbean. These are some of the slowest-growing and longest-lived reef-building corals. Fast-spreading disease is a 'potential global threat'.

Statement 1 is correct: The National Oceanic and Atmospheric Administration plan aims to halt the spread of stony coral tissue loss disease as this a newly discovered **affliction confirms 94% mortality rate among almost two dozen coral species**.

Statement 2 is correct: The disease was identified for the first time in Florida eight years ago, the disease has already decimated large sections of Florida's Coral Reef, North America's only barrier reef, and quickly spread to the U.S. Virgin Islands, Puerto Rico and at least 20 countries in the Caribbean region.

A key element of the National Oceanic and Atmospheric Administration plan is preventing its spread to the Pacific Ocean. NOAA is working with several Pacific nations on a possible collective response to the disease's migration to other oceans, which is regarded as nearly inevitable.

X Factor:

Scientists found evidence the disease recently invaded the **Gulf of Mexico**. The disease's rapid spread and coral-killing prowess are unprecedented.

NOAA is shipping small colonies of the sanctuary's corals to an aquarium in Galveston, Texas. The rescued corals could be used to repopulate areas hit hard by the disease.

Source:

https://www.nola.com/news/environment/article_56bedef8-465b-11ed-b33b-cbd9abc546d3.html

60. (a)

Explanation:

Some of the important passes of the Kashmir region are Zoji La on the Great Himalayas, Banihal on the Pir Panjal, Photu La on the Zaskar and Khardung La on the Ladakh range. Some of the important fresh lakes such as Dal and Wular

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and salt water lakes such as Pangong Tso and Tso Moriri are also in this region.

Pair 1 is correct: Zoji La is a high mountain pass at an elevation of 3,528m (11,575ft) above the sea level, located in the Kargil district of Ladakh, in India: **Zozila pass** is located in Ladakh, **Great Himalayas Range**.

Pair 2 is incorrect: Khardung la pass is located in Ladakh Range. Famously known as the gateway to the Nubra and Shyok Valleys in the Ladakh region of Jammu and Kashmir.

Pair 3 is incorrect: Photu la is a mountain pass on the Srinagar-Leh highway in the **Zaskar Range of the Himalayas** in India. At an elevation of 4,108 metres (13,478 ft), it is the highest point on the highway, surpassing the famed Zoji La

Pair 4 is incorrect: Banihal pass is located in **Pir Pangal** range, it connects Jammu and Shrinagar.

X Factor

- Nathu La pass located in the state of Sikkim and also in India-China border
- Shipki la pass is located through Sutlej Gorge, connects Himanchal Pradesh with Tibet.
- Niti Pass connects Uttarakhand with Tibet.
- Rohtang Pass (Kullu -Lahul-Spiti) located in the state of Himanchal Pradesh.

Source: Book : NCERT Class 11 , India Physical Environment Geography Chapter 2 ,Structure and Physiography

Source:

<https://education.nationalgeographic.org/resource/pass>

61. (b)

Explanation:

The Darjeeling and Sikkim Himalaya:

Statement 1 is correct The British, taking advantage of the physical conditions such as moderate slope, thick soil cover with high organic content, well distributed rainfall throughout the year and mild winters, introduced tea plantations in this region. As compared to the other sections of the Himalayas, these along with the Arunachal Himalayas are conspicuous by the absence of the Shiwalik formations. **In place of the Shiwaliks here, the 'duar formations' are**

important, which have also been used for the development of tea gardens.

Statement 2 is incorrect: Duar formations are characteristic feature of the **Darjeeling and Sikkim Himalayas**. In place of shasliks these hills have duar formations.

Statement 3 is incorrect: The two distinguishing features of The Himachal and Uttarakhand Himalayas region from the point of view of physiography are the 'Shiwalik' and 'Dun' formations. Some important duns located in this region are the Chandigarh-Kalka dun, Nalagarh dun, Dehra Dun, Harike dun and the Kota dun, etc. Dehra Dun is the largest of all the duns with an approximate length of 35-45 km and a width of 22-25 km. **The famous 'Valley of flowers' is also situated in this region. Valley of flower is example of dun formation.**

X factor

- Valley of Flowers National Park is a UNESCO World Heritage Site and forms one of the two core zones (the other being the Nanda Devi National Park) of the Nanda Devi Biosphere Reserve.

Source: NCERT Class 11, India Physical Environment Geography page no.13

<https://www.britannica.com/place/Duars>

<https://www.euttaranchal.com/tourism/valley-of-flowers.php>

62. (b)

Explanation:

Option (b) is correct:

Darjiling and Sikkim Himalayas

The two distinguishing features of this region from the point of view of physiography are the 'Shiwalik' and 'Dun' formations. Some important duns located in this region are the Chandigarh-Kalka dun, Nalagarh dun, Dehra Dun, Harike dun and the Kota dun, etc. Dehra Dun is the largest of all the duns with an approximate length of 35-45 km and a width of 22-25 km. In the Great Himalayan range, the valleys are mostly inhabited by the Bhotia's. These are nomadic groups who migrate to **'Bugyals' (the summer grasslands in the higher reaches)** during summer months and return to the valleys during winters. The famous

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'Valley of flowers' is also situated in this region. The places of pilgrimage such as the Gangotri, Yamunotri, Kedarnath, Badrinath and Hemkund Sahib are also situated in this part.

Source: NCERT Class 11, India Physical Environment Geography Page No. 20

63. (d)

Explanation:

Option (d) is correct:

The Kashmir Himalayas are also famous for Karewa formations, which are useful for the cultivation of Zafran, a local variety of saffron. **Some of the important passes of the region are Zoji La on the Great Himalayas, Banihal on the Pir Panjal, Photu La on the Zaskar and Khardung La on the Ladakh range.** Some of the important fresh lakes such as Dal and Wular and **salt water lakes such as Pangong Tso and Tso Moriri are also in this region.**

X factor:

- **Pangong Tso:** saline water lake located at an altitude of 4350 m above the sea level. However, even though it is a **saline water lake**, Pangong Tso freezes completely during winters. Pangong Tso is popular for its changing colour ability.
- **Tso Moriri lake in Changthang region of Ladakh** is the largest high-altitude lake in India that is entirely in Indian Territory. This is **world's highest salt water lake at 14836 ft.**
- Being part of the wetland reserve under Ramsar site, it is actual known as **Tsomoriri Wetland Conservation Reserve** and one cannot pitch any tents or construct anything near the banks of the lake.

Source: NCERT Class 11, India Physical Environment Geography Page No. 11.

<https://leh.nic.in/tourist-place/tsomoriri-lake/>

<https://www.news18.com/news/india/the-last-run-a-marathon-over-ladakhs-frozen-pangong-lake-to-draw-attention-to-melting-himalayan-glaciers-5712535.html>

64. (c)

Explanation:

Option (c) is correct

Eastern Ghats:

The Eastern Ghats, once covered by luxuriant forests, are becoming barren because of the greed and endless quest of mankind in the name of development. The series of broken hills run from the Mahanadi in Odisha to the Vaigai in Tamil Nadu, with an average elevation of 1,000 metres in peninsular India.

Small Rivers Flowing towards the East: There are a large number of rivers flowing towards the east along with their tributaries. There are small rivers which join the Bay of Bengal, though small, these are important in their own right. The **Subarnrekha, the Baitarni, the Brahmani, the Vamsadhara, the Penner**, the Palar and the Vaigai are important rivers.

West flowing rivers:

The **Mandovi River is one of the main West Flowing Rivers of Goa State.** The river runs in the North- East direction for about 5 km and then follows in the West ward direction.

The Mandovi River rises in the Jamboti Ghats in Karnataka State. At the origin, near the village Mabulyesheir, it is known as Bhaburnal, which is at an elevation of 600m above M.S.L. The total length of this West Flowing River from its origin to its out fall into the Arabian Sea is 62 km.

The Vaitarna rises from the Trimbak hills in Nasik district at an elevation of 670 m. The Kalinadi rises from Belgaum district and falls in the Karwar Bay. The source of Bedti river lies in Hubli Dharwar and traverses a course of 161 km. **The Sharavati is another important river in Karnataka flowing towards the west.** The Sharavati originates in Shimoga district of Karnataka and drains a catchment area of 2,209 sq. km

Source: India: physical environment X1, Page No. 28-29

<http://www.cwc.gov.in/sites/default/files/admin/West-Flowing-Rivers-Basin-Write-2008-09.pdf>

<https://www.downtoearth.org.in/blog/wildlife-biodiversity/eastern-ghats-a-biota-under-serious-threat-63456>

65. (d)

Explanation:

TEST – 9009 (EXPLANATION & SOURCE)

Option (d) is correct

The Intertropical Convergence Zone, or ITCZ, is the region that circles the Earth, near the equator, where the trade winds of the Northern and Southern Hemispheres come together.

- The intense sun and warm water of the equator heats the air in the ITCZ, raising its humidity and making it buoyant. Aided by the convergence of the trade winds, the buoyant air rises. As the air rises it expands and cools, releasing the accumulated moisture in an almost perpetual series of thunderstorms.
- Seasonal shifts in the location of the ITCZ drastically affects rainfall in many equatorial nations, resulting in the wet and dry seasons of the tropics rather than the cold and warm seasons of higher latitudes. Longer term changes in the ITCZ can result in severe droughts or flooding in nearby areas.
- The Inter Tropical Convergence Zone (ITCZ,) is a broad **trough of low pressure in equatorial latitudes**. This is where the northeast and the southeast trade winds converge
- This convergence zone lies more or less **parallel to the equator but moves north or south with the apparent movement of the sun**

The shift of the position of Inter Tropical Convergence Zone (ITCZ) in summer, over the Ganga plain (this is the equatorial trough normally positioned about 5°N of the equator effects monsoon. It is also known as the monsoon trough during the monsoon season.

X factor

- The front where the south-west monsoons meet the north-east trade winds is known as the Monsoon Front (ITCZ) thus rainfall occurs along this front.
- In July the ITCZ shifts to 20°- 25° N latitude and is located in the Indo-Gangetic Plain and the south-west monsoons blow from the Arabian Sea and the Bay of Bengal, this is called as Monsoon Trough.
- The seasonal shift of the ITCZ has given the concept of Northern Inter-Tropical Convergence Zone (NITCZ) in summer

(July – rainy season) and Southern Inter-Tropical Convergence Zone (SITCZ) in winter (Jan – dry season).

Source: NCERT Class 11, India Physical Environment Geography Chapter 4, Climate <https://earthobservatory.nasa.gov/images/703/the-intertropical-convergence-zone>

66. (d)

Explanation:

Option (d) is correct

El Niño is a complex weather pattern that occurs when surface water in the central and eastern Pacific Ocean becomes warmer than average and east winds blow weaker than normal. The events, which typically occur every few years, can produce unusual and, sometimes dangerous, weather conditions around the world including droughts, floods and severe storms.

- **El Nino and La Nina are the warm and cool phases** of a recurring climate pattern across the tropical Pacific—the El Nino-Southern Oscillation, or “ENSO”.
- **El Niño favours stronger hurricane activity in the central and eastern Pacific basins, and suppresses it in the Atlantic basin** on the other hand La Niña suppresses hurricane activity in the central and eastern Pacific basins, and enhances it in the Atlantic basin.
- The Atlantic basin, the amplified trough is associated with stronger upper-level westerly winds and stronger lower-level easterly trade winds, both of which increase the vertical wind shear and suppress hurricane activity. To enhanced vertical wind shear, **El Nino suppresses Atlantic hurricane** activity by increasing the amount of sinking motion and increasing the atmospheric stability.
- This rapid melting of Arctic ice is not just disruptive to surrounding coastal cities and small island nations; it also may have a lasting impact on global weather patterns. **Researchers have found out that if Arctic becomes seasonally ice-free, the frequency of strong El Niño events increases significantly.**

Source: NCERT Class 11, India Physical Environment Geography Chapter 4, Climate

TEST – 9009 (EXPLANATION & SOURCE)

<https://phys.org/news/2022-09-arctic-sea-ice-loss-frequent.html>

<https://www.climate.gov/news-features/blogs/enso/impacts-el-ni%C3%B1o-and-la-ni%C3%B1a-hurricane-season>

67. (c)

Explanation:

According to demographic population of any region changes from high births and high deaths to low births and low deaths as society progresses from rural agrarian and illiterate to urban industrial and literate society. These changes occur in stages which are collectively known as the demographic cycle.

Pair 1 is incorrect: The first stage has **high fertility and high mortality** because people reproduce more to compensate for the deaths due to epidemics and variable food supply

Pair 2 is correct: Fertility remains high in the beginning of second stage but it declines with time. This is accompanied by **reduced mortality rate**. Improvements in sanitation and health conditions lead to decline in mortality.

Pair 3 is incorrect: In the last stage, **both fertility and mortality decline considerably**. The population is either stable or grows slowly. The population becomes urbanized, literate and has high technical knowhow and deliberately controls the family size.

X-Factor:

- The annual population growth rate in India is 1.64 per cent.
- Developed countries take more time to double their population as compared to developing.
- It took more than a million years for the human population to attain the one billion marks. But it took only 12 years for it to rise from 5 billion to 6 billion.

Source: NCERT 12th: Fundamentals of Human Geography, Chapter 2, Page no. 14

68. (c)

Explanation:

Option (c) is correct

- **Violent thunderstorms in the Gangetic plains of India are locally known as Kal Baisakhi or Nor'westers.** These localised events are generally associated with thunderstorms accompanied by strong squally winds and torrential rainfall.
- Kal Baisakhi originates in the Chhotanagpur Plateau, in and around the cities of Ranchi and Jamshedpur. In times of incursion of moist air in these elevated regions, amidst high temperatures, these extremely destructive thunderstorms occur.
- **Kal Baisakhi originates over Bihar and Jharkhand area, moves eastwards and strikes West Bengal and Odisha, takes Chhattisgarh under its purview as well.**
- Kal Baisakhi definitely brings destruction in terms of lightning, thunderstorm, hailstorm and rainfall. At the same time, it is extremely helpful for the pre-Kharif crops like jute, paddy and a large number of vegetables and fruits. It gives the much-desired relief after mid-day heat and pours well on the thirsty soil for development of crops.
- Kal Baisakhi is a common occurrence during April and May and few events of Kal Baisakhi are seen in March too.

Bardoli Chheerha:

The localised thunderstorms, associated with violent winds, torrential downpours, often accompanied by hail are referred to as Bardoli in Assam

Blossom Shower: The pre -monsoon storms are refer to as Blossom Shower in Kerala.

X factor:

Some names for pre monsoon in different states:

- **Tea showers** in Assam as they are good for tea, jute and rice.
- **Mango showers** in Kerala and coastal areas of Karnataka, they help in the early ripening of mangoes.
- **Cherry Blossoms/ Coffee showers** in Kerala and nearby areas .They for good for coffee plantations)

Source: NCERT Class 11, India Physical Environment Geography Chapter 4, Climate

<https://www.skymetweather.com/content/weather-faqs/what-is-kal-baisakhi-or-norwesters/>

TEST – 9009 (EXPLANATION & SOURCE)

69. (b)

Explanation:

Statements 1 and 4 are correct:

- The oldest female is the leader of the herd, known as matriarch. A herd may have 10 to 12 female elephants and young ones. Male elephants live in the herd till they are 14–15 years old. Then they leave their herd and move around alone
- **Elephants have the longest gestation period of all mammals, carrying their young for 18 to 22 months before giving birth.**
- Long developmental periods are common among highly intelligent animals. Since elephants are the largest living and biggest-brained land animals in the world, there's a lot of development for elephants to do in the womb.

Statements 2 and 3 are incorrect:

- **Asian Elephants are considered endangered species** by IUCN red list while African Forest Elephant are considered critically endangered.
- As per the Elephant census (2017) **Karnataka has the highest number of elephants followed by Assam and Kerala.**

X Factor:

Some important projects related to elephant conservation:

- Project Elephant (1992)
- Monitoring the Illegal Killing of Elephants (MIKE) programme, launched in 2003; The MIKE Programme was established by the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES).
- Gaj Yatra
- Field Manual for Managing Human-Elephant Conflicts (HEC) by Ministry of Environment, Forest and Climate Change at 16th Steering Committee meeting of **Project Elephant**

Source: NCERT Class 11, India Physical Environment Geography Chapter 5, Natural Vegetation

Source:

<https://www.naturesafariindia.com/wildlife-in-india/project-elephant/>

70. (a)

Explanation:

Statement 1 is incorrect: United Nations' World Population Prospects (WPP) has been publishing the World Population Report in a biennial cycle since 1951. Each revision of the WPP provides a historical time series of population indicators starting in 1950. It does so by taking into account newly released national data to revise estimates of past trends in fertility, mortality or international migration.

Statement 2 is incorrect: The report said **global population growth fell below 1% in 2020 for the first time since 1950.** According to the latest U.N. projections, the world's population could grow to around 8.5 billion in 2030, 9.7 billion in 2050 and a peak of around 10.4 billion during the 2080s. It is forecast to remain at that level until 2100.

Statement 3 is correct: As for gender balance, the report says, "Globally, the world counts slightly more men (50.3%) than women (49.7%) in 2022." "This figure is projected to slowly invert over the course of the century," it says. **"By 2050, it is expected that the number of women will equal the number of men."**

X-factor:

Other highlights of the report:

- The world's population will reach 8 billion on Nov. 15 and that India will replace China as the world's most populous nation next year.
- The report projects that next year India, with a current population of 1.412 billion, will surpass China, with a current population of 1.426 billion.
- The U.N. projects that in 2050 the United States will remain the third most populous country in the world, behind India and China. Nigeria is projected to be No. 4, followed by Pakistan, Indonesia, Brazil, Congo, Ethiopia and Bangladesh. Russia and Mexico, which are in the top 10 most populous countries in 2022, are projected to lose their ninth and 10th spots in 2050.
- The share of working age people between ages 25 and 64 has been increasing in most countries of sub-Saharan Africa,

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parts of Asia and in Latin America and the Caribbean “thanks to recent reductions in fertility,” the report says.

Source:

<https://indianexpress.com/article/world/un-projects-world-population-will-reach-8-billion-on-nov-15-8024001/>

<https://indianexpress.com/article/explained/reading-un-population-report-8023378/>

71. (a)

Explanation:

Statements 1 is correct but 2 is incorrect:

Distribution of salinity: On an average the **salinity decreases from equator towards the poles.**

The **highest salinity is observed between 20° and 40° north latitudes** because this zone is characterized by **high temperature, high evaporation** but **less rain than the equatorial region.**

The marginal areas of the oceans bordering the continents have lower salinity than their interior due to addition of fresh water to the marginal areas through the rivers.

X factor:

Very high salinity is recorded in Lake Von, Turkey (330‰) Dead Sea (238‰) and Great Salt Lake, Utah, USA (220‰).

Source: SCERT: TN: XI: Geography: Chapter: 5: Hydrosphere: Page:122-123.

72. (c)

Explanation:

Option(c) is correct:

When in **geostrophic balance**, wind in the atmosphere has a balance between the pressure gradient force (PGF) and the Coriolis force (CF). In geostrophic balance, $PGF = CF$. The resulting wind is called a geostrophic wind.

Jet streams are some of the geostrophic winds, flowing in the atmosphere. Their speeds usually range from 129 to 225 kilometers per hour (80 to 140 miles per hour), but they can reach more than 443 kilometers per hour (275 miles per hour). They are faster in winter when the temperature differences between tropical, temperate, and polar air currents are greater. **High speed of wind at**

upper atmosphere due to lack of friction increase coriolis is which balances the PGF and flow as Geostrophic wind.

X-Factor:

The PGF is the force resulting in the pressure difference between higher and lower pressure. The PGF points directly from high to low pressure (or heights). The Coriolis force results from the spin of the Earth and is directed to the right of the path of motion in the Northern Hemisphere

Source: Fundamental of physical geography, NCERT page no. 91

<https://education.nationalgeographic.org/resource/jet-stream>

<https://www.theweatherprediction.com/habyhints3/741/>

73. (b)

Explanation:

There are many ways of looking at the problem of human development. Some of the important approaches are: (a) The income approach; (b) The welfare approach; (c) Minimum needs approach

Statement 1 is incorrect: Basic Needs Approach was initially proposed by the International Labour Organisation (ILO). Six basic needs i.e.: **health, education, food, water supply, sanitation, and housing** were identified. The question of human choices is ignored and the emphasis is on the provision of basic needs of defined sections.

Statement 2 is correct: Welfare Approach looks at human beings as beneficiaries or targets of all development activities. The approach argues for higher government expenditure on education, health, social secondary and amenities. **People are not participants in development but only passive recipients.** The government is responsible for increasing levels of human development by maximizing expenditure on welfare.

Statement 3 is correct: Capability Approach is associated with **Prof. Amartya Sen. Building human capabilities in the areas of health, education and access to resources is the key to increasing human development.**

TEST – 9009 (EXPLANATION & SOURCE)

Source: NCERT 12th: Fundamentals of Human Geography, Chapter 4, Page no. 26

74. (b)

Explanation:

Statement 1 is correct: The Total Fertility Rate (TFR) -- which is the average number of children who would be born to any women in her lifetime – has declined from 2.2 in 2015-16 to 2.0 in 2019-21.

Statement 2 is correct: India's TFR of two is currently below the replacement level of fertility of 2.1 children per woman.

Replacement level fertility represents the level at which a population exactly replaces itself from one generation to the next, thus leading to zero population growth if the level sustained over a sufficiently long period

Statement 3 is incorrect: The TFR among women in rural areas has declined from 3.7 children in 1992-93 to 2.1 children in 2019-21 i.e. a decline of 1.6

The corresponding decline among women in urban areas was from 2.7 children in 1992-93 to 1.6 children in 2019-21 i.e. a decline of 1.1

X-factor:

- In all NFHS surveys, irrespective of place of residence, the fertility rate peaks at the age of 20-24, after which it declines steadily.
- Another interesting aspect was that the number of children per woman declines with women's level of schooling. Women with no schooling have an average of 2.8 children, compared with 1.8 children for women with 12 or more years of schooling.
- The TFR ranges from 1.1 children per woman in Sikkim to three children per woman in Bihar. Five states are yet to achieve a replacement-level of fertility of 2.1, according to the NFHS-5.
- These states are Bihar (2.98), Meghalaya (2.91), Uttar Pradesh (2.35), Jharkhand (2.26) and Manipur (2.17).
- All the communities are witnessing a decline in fertility. The TFR for Hindu women was at 1.94, and for Muslim women it was 2.2. The Christian community has a fertility rate of 1.88, the Sikh community 1.61, the Jain

community 1.6 and the Buddhist and neo-Buddhist community 1.39.

Source: https://www.business-standard.com/podcast/current-affairs/india-s-fertility-rate-dips-below-replacement-level-what-does-it-mean-122051800062_1.html

75. (c)

Explanation:

Option(c) is correct:

Statement 1 is correct: The air containing moisture to its full capacity at a given temperature is said to be saturated.

It means that the air at the given temperature is incapable of holding any additional amount of moisture at that stage.

Statement 2 is correct: The temperature at which saturation occurs in a given sample of air is known as dew point.

Source: NCERT, Fundamentals of physical geography Page No 99.

76. (c)

Explanation:

Statement 1 is incorrect: Sulu Sea, portion of the western North Pacific Ocean. It is bounded by northeastern Borneo on the southwest, the southwestern islands of the Philippines, including Palawan, on the west and northwest, Busuanga and Mindoro on the north, Panay and Negros on the east, and Mindanao and the Sulu Archipelago on the southeast.

Statement 2 is correct: A fracture line, bisecting the sea from northeast to southwest, is evidenced by the coral atolls surrounding Mapun (Cagayan Sulu) island, the Tubbataha Reefs, and **the volcanic Mapun island group** itself.

Statement 3 is correct: The Tubbataha Reefs were designated a UNESCO World Heritage site in 1993 in recognition of their abundance and diversity of marine life; in 2009 the boundaries of the site were extended to triple its original size.

X-factor:

Northern Atlantic Ocean:

- The Sargasso Sea, located entirely within the Atlantic Ocean, is the **only sea without a land boundary**.

TEST – 9009 (EXPLANATION & SOURCE)

- The Sargasso Sea is a vast patch of ocean named for a genus of free-floating seaweed called *Sargassum*.
- It lies within the Northern Atlantic Subtropical Gyre.

Source:

<https://www.britannica.com/place/Celebes-Sea>

<https://indianexpress.com/article/technology/science/chinese-space-rocket-debris-earth-indian-ocean-8061811/>

<https://oceanservice.noaa.gov/facts/sargassosea.html>

77. (a)

Explanation:

The mesosphere lies above the stratosphere, which extends up to a height of 80 km. In this layer, once again, temperature starts decreasing with the increase in altitude and reaches up to minus 100° C at the height of 80 km. The upper limit of mesosphere is known as the mesopause.

The Ionosphere is part of Earth's upper atmosphere, between 80 and about 600 km where Extreme UltraViolet (EUV) and x-ray solar radiation ionizes the atoms and molecules thus creating a layer of electrons. The ionosphere is important because it reflects and modifies radio waves used for communication and navigation.

Statement 1 is correct: Radio waves transmitted from the earth are reflected back to the earth by ionosphere.

Statement 2 is incorrect: Temperature here starts increasing with height.

Source: NCERT, Fundamentals of physical geography Page No. 77

78. (c)

Explanation:

Pair 1 is incorrectly matched: The Danube Waterway: This important inland waterway serves Eastern Europe. The Danube River rises in the Black Forest and flows eastwards through many countries. It is navigable up to Taurna Severin. The chief export items are wheat, maize, timber, and machinery.

Pair 2 is correctly matched: The Volga Waterway: Russia has a large number of developed waterways, of which the Volga

is one of the most important. It provides a navigable waterway of 11,200 km and drains into the Caspian Sea. The Volga-Moscow Canal connects it with the Moscow region and the Volga-Don Canal with the Black Sea.

Pair 3 is correctly matched: The Mississippi Waterways: The Mississippi-Ohio waterway connects the interior part of U.S.A. with the Gulf of Mexico in the south. Large steamers can go through this route up to Minneapolis.

Pair 4 is correctly matched: The Rhine Waterways: The Rhine flows through Germany and the Netherlands. It is navigable for 700 km from Rotterdam, at its mouth in the Netherlands to Basel in Switzerland. Ocean-going vessels can reach up to Cologne. The Ruhr river joins the Rhine from the east. It flows through a rich coalfield and the whole basin has become a prosperous manufacturing area. Dusseldorf is the Rhine port for this region.

X-factor:

Shipping Canals: The Suez and the Panama Canals are two vital man-made navigation canals or waterways which serve as gateways of commerce for both the eastern and western worlds.

- **The Suez Canal:** This canal had been constructed in 1869 in Egypt between Port Said in the north and Port Suez in the south linking the Mediterranean Sea and the Red Sea. It gives Europe a new gateway to the Indian Ocean and reduces direct sea-route distance between Liverpool and Colombo compared to the Cape of Good Hope route. It is a sea-level canal without locks which is about 160 km and 11 to 15 m deep. About 100 ships travel daily and each ship takes 10-12 hours to cross this canal. The tolls are so heavy that some find it cheaper to go by the longer Cape Route whenever the consequent delay is not important.
- **The Panama Canal:** This canal connects the Atlantic Ocean in the east to the Pacific Ocean in the west. It has been constructed across the Panama Isthmus between Panama City and Colon by the U.S. government which purchased 8 km of area on either side and named it the Canal Zone. The Canal is about 72 km long.

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Source: Human Geography 12th NCERT; Chapter-8; Page- 74,76,77

79. (c)

Explanation:

Statement 1 is correct: Anthrax is a zoonotic disease primarily affecting ruminant herbivores such as cattle, sheep, goats, antelope, and deer that become infected by ingesting contaminated vegetation, water, or soil; humans are generally incidental hosts.

Statement 2 is incorrect: Anthrax is most common in agricultural regions in Central and South America, sub-Saharan Africa, central and southwestern Asia, and southern and eastern Europe. Although outbreaks still occur in livestock and wild herbivores in the United States, Canada, and Western Europe.

Statement 3 correct: Anthrax is a serious infectious disease caused by gram-positive, rod-shaped **bacteria known as *Bacillus anthracis***. It occurs naturally in soil and commonly affects domestic and wild animals around the world.

People can get sick with anthrax if they come in contact with infected animals or contaminated animal products. Anthrax can cause severe illness in both humans and animals. Anthrax is **not** contagious, which means you can't catch it from another person like the cold or flu.

Statement 4 is correct: The **standard treatment for anthrax** is an antibiotic such as **ciprofloxacin (Cipro), doxycycline (Vibramycin).**

X-Factor:

- Anthrax is most common in developing countries but can also be found globally.
- The largest reported agricultural outbreak occurred in Zimbabwe, with more than 10,000 cases reported between 1979 and 1985.

- **Human anthrax in India is a notifiable disease.** It commonly affects communities living near forest areas and outbreaks are predominantly reported from eastern states such as Odisha.

Source:

<https://www.cdc.gov/anthrax/basics/index.html>

<https://wwwnc.cdc.gov/travel/yellowbook/2020/travel-related-infectious-diseases/anthrax>

<https://www.nhp.gov.in/disease/anthrax>

<https://rarediseases.org/rare-diseases/anthrax/>

80. (b)

Explanation:

Option (b) is correct:

The expert committee is related to **Road Safety and Traffic Management in 2005**. It recommended the creation of the National Road Safety & Traffic Management Board, an Apex body at national level to promote road safety and traffic management in the country to be constituted through an Act of the Parliament with members and experts drawn from the various fields including road engineering, automobile engineering, traffic laws, medical care, etc.

X-Factor:

- According to the latest NCRB report, deaths by accidents on roads increased by almost 17 per cent in 2021 compared to 2020, indicating an increase in the rate of deaths per 1,000 vehicles in India.
- Two-wheelers accounted for most deaths (44.5%). Buses accounted for 3% of deaths in accidents. Speeding caused 87,000 deaths, accounting for over half of all deaths, while dangerous and careless driving was attributed as cause for 42,000 deaths.

Source: <https://indianexpress.com/article/upsc-current-affairs/upsc-essentials/case-study-of-the-week-road-safety-better-late-than-never-8140872/>
<https://indianexpress.com/article/india/nrb-report-deaths-in-road-accidents-up-by-17-pc-8119743/>

81. (d)

Explanation:

The Polar Satellite Launch Vehicle (PSLV) is the third generation launch vehicle of India. It is the first Indian launch vehicle to be equipped with liquid stage. It is an expendable launch vehicle which can place 1,600kg remote sensing satellites into sun-synchronous orbit (SSO) and 1,050kg small-sized spacecraft into geostationary transfer orbit (GTO).

www.ksgindia.com

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Statement 1 is correct: PSLV-C53 is the second dedicated commercial mission of New Space India Limited (NSIL). This is the 55th mission of PSLV and 15th mission using PSLV-Core Alone variant. It is the 16th PSLV launch from the second launch pad. The mission proposes to demonstrate the utilization of the spent upper stage of the launch vehicle as a stabilized platform for scientific payloads subsequent to the separation of the satellites.

Statement 2 is correct: ISRO uses a "stabilised" fourth stage of the PSLV launch vehicle to conduct in-orbit experiments. This — **PSLV Orbital Experimental Module (POEM)** — will be tested in the PSLV-C53 mission. POEM will demonstrate the utilization of the spent upper stage of the launch vehicle as a stabilized platform for scientific payloads subsequent to the separation of the satellites. **POEM activity performs in-orbit scientific experiments using the spent PS4 (fourth) stage as an orbital platform. It is the first time that the PS4 stage would orbit the earth as a stabilized platform.**

Statement 3 is correct: PSLV-C53 carries three satellites. DS-EO, a 365 kg and NeuSAR, a 155 kg satellite both belonging to Singapore. Third satellite is a 2.8 kg Scoob-1 of Nanyang Technological University (NTU), Singapore. DS-EO carries an Electro-Optic, multi-spectral payload that will provide full color images for land classification, and serving Humanitarian Assistance and Disaster Relief needs. NeuSAR is Singapore's first small commercial satellite carrying a SAR payload, which is capable of providing images in day and night and under all weather conditions. SCOOB-I satellite is the first satellite in the Student Satellite Series (S3-I), a hands-on student training, in Sun-Synchronous Polar Orbits

X-Factor:

- The Indian Space Research Organisation (ISRO) is developing a rocket named Next Generation Launch Vehicle (NGLV) to replace its ageing workhorse the Polar Satellite Launch Vehicle (PSLV).
- It will use 'semi-cryogenic' technology which is both efficient and cost-effective.

Source:https://www.isro.gov.in/mission_PSLV_C53.html

<https://timesofindia.indiatimes.com/india/in-a-1st-pslv-4th-stage-to-be-used-to-conduct-in-orbit-experiments-june-30-mission-to-launch-3-singapore-satellites/articleshow/92394574.cms>

<https://www.thehindu.com/sci-tech/science/pslv-c-53-carrying-three-commercial-satellites-lifts-off/article65586051.ece>

<https://www.newindianexpress.com/states/kerala/2022/oct/13/isro-developing-new-rocket-to-replace-pslv-2507752.html>

82. (a)

Explanation:

Statement 1 is incorrect: Sea of Marmara lies between Thrace and Anatolia parts of Turkey, covering an area of over 11 thousand square kilometers (4247 square miles). It is approximately 280 kilometers (174 miles) long from northeast to southwest and about 80 kilometers (50 miles) wide at its greatest width. Its maximum depth reaches 1355 meters (4445 feet) near the center.

Statement 2 is correct: Sea of Marmara is an inland sea within the Marmara region connecting to the Black Sea with the Bosphorus Strait in the northeast, and to the Aegean Sea with the Dardanelles Strait in the southwest.

Statement 3 is incorrect: Zmiinyi Island, also known as Snake or Serpent Island, is a small piece of rock less than 700 metres from end to end, that has been described as being "X-shaped". It is located 35 km from the coast in the Black Sea, to the east of the mouth of the Danube and roughly southwest of the port city of Odessa.

X-Factor:

- Spread across an area of 436,400 km², the Black is located in Eurasia, surrounded by Europe, Caucasus and Anatolia. The countries that share a border with the Black Sea include Romania, Turkey, Bulgaria, Ukraine, Russia, and Georgia.
- It is surrounded by the Pontic, Caucasus, and Crimean Mountains in the south, east and north, respectively. The

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Strandzha mountains also constrain it to the southwest and Dobrogea Plateau to the northwest.

- It is anoxic water. Precisely, there is a significant absence of oxygen in the water. The Black Sea is the largest water body with a meromictic basin, which means the movement of water between the lower and upper layers of the sea is a rare phenomenon to find anywhere in the world.

Source: <https://indianexpress.com/article/explained/everyday-explainers/explained-importance-snake-island-black-sea-ukraine-bombed-russia-7985298/>

<https://www.allaboutturkey.com/sea.html>

<https://www.marineinsight.com/know-more/8-amazing-facts-about-the-black-sea/>

83. (b)

Explanation:

The Kaveri is one of the major rivers flowing through Karnataka. As it descends from the Deccan plateau and makes its way to the Arabian sea, it forms a number of tributaries. The river Harangi is one such tributary

Statement 1 is correct: Tributaries of the Cauvery River:

Left Bank: **The Harangi**, the Hemavati, the Shimsha and the Arkavati.

Right Bank: Lakshmantirtha, the Kabbani, the Suvarnavati, the Bhavani, the Noyil and the Amaravati joins from right.

Statement 2 is correct: The Harangi river originates in the Pushpagiri hills of the Western Ghats and being the only reservoir in the region of Coorg, and it holds great tranquil and picturesque value. The **Dubare forest** area covers the place with its verdant trees and grasslands.

Statement 3 is incorrect: Harangi Dam is about 36 km from Madikeri town. The construction of Harangi Dam started in the 1970s and was opened for operation in the year 1982. The height of the dam 53 m and the boundary of the dam stretches for 845 m. It is a large dam; the reservoir of the dam is spread across the area of 420 sq km. It is a hydroelectric and irrigation project, which

was built to produce hydropower and it is also used for irrigation purposes in the nearby areas. **Amaravathi Dam, on Amaravathi River (a right bank tributary of Cauvery) is located in Indira Gandhi Wildlife Sanctuary in Tiruppur district, Tamil Nadu, India.**

X-Factor:

- Kaveri river rises on Brahmagiri Hill of the Western Ghat in southwestern Karnataka state, flows in a southeasterly direction for 475 miles (765 km) through the states of Karnataka and Tamil Nadu, and descends the Eastern Ghats in a series of great falls.
- Before emptying into the Bay of Bengal south of Cuddalore, Tamil Nadu, the river breaks into a large number of distributaries forming a wide delta called the “**garden of southern India.**” Known to devout Hindus as Daksina Ganga (“Ganges of the South”)

Source:

<https://www.karnataka.com/coorg/river-harangi/>

<https://coorgtourism.co.in/harangi-dam-coorg>

<https://www.explorebees.com/Madikeri/attraction/Harangi-Reservoir/pl85516>

<https://www.britannica.com/place/Kaveri-River>

84. (a)

Explanation:

tropical cyclone, also called **typhoon** or **hurricane**, an intense circular storm that originates over warm tropical oceans and is characterized by low atmospheric pressure, high winds, and heavy rain.

Statement 1 and 2 are correct: Tropical cyclones draw energy from warm ocean surfaces and use this energy to strengthen a cyclonic disturbance into an intense vortex, which is called a hurricane, typhoon, or cyclone, depending on where in the world it occurs. The tropical cyclone initiates from a collection of cumulonimbus clouds before the intense vortex dominates the dynamics. However, the convective clouds in the genesis stage exhibit differences from ordinary cumulonimbus, including very intense “convective bursts” and rotating convective cells called “vortical hot towers.”

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Statement 3 is incorrect: Generally, it is formed in tropic seas away from equator/equatorial region due to absence of coriolis force.

Every year during the late summer months (**July–September in the Northern Hemisphere and January–March in the Southern Hemisphere**), cyclones strike regions as far apart as the Gulf Coast of North America, northwestern Australia, and eastern India and Bangladesh.

Statement 4 is incorrect: Tropical cyclones are known by various names in different parts of the world. In the North Atlantic Ocean and the eastern North Pacific they are called hurricanes, and in the western North Pacific around the Philippines, Japan, and China the storms are referred to as typhoons. **In the western South Pacific and Indian Ocean they are variously referred to as severe tropical cyclones, tropical cyclones, or simply cyclones.** All these different names refer to the same type of storm.

Noru, known locally as Karding, a super typhoon, **occurred in Philippines.**

X-Factor:

- The Indian subcontinent is one of the worst affected regions in the world.
- The subcontinent with a long coastline of 8041 kilometres is exposed to nearly 10 per cent of the world's tropical cyclones.
- Majority of them have their initial genesis over the Bay of Bengal and strike the East coast of India
- More cyclones occur in the Bay of Bengal than the Arabian Sea and the ratio is approximately 4:1. Cyclones occur frequently on both the coasts (the West coast - Arabian Sea; and the East coast - Bay of Bengal).

Source:

<https://www.britannica.com/science/tropical-cyclone>
<https://www.bbc.com/news/world-asia-63023891>
<https://ndma.gov.in/Natural-Hazards/Cyclone>

85. (b)

Explanation:

Statement 1 is incorrect: Nitrogen is one of the most **important macronutrients** needed

for development of a plant. **it plays a fundamental role in energy metabolism and protein synthesis.** Nitrogen is absorbed by the plant in the form of a nitrate. This macronutrient is directly related to plant growth. It is indispensable for photosynthesis activity and chlorophyll formation.

Statement 2 is correct and statement 3 is incorrect: It is a **part of chlorophyll, amino acids and nucleic acids**, among others. Such organisms are called "diazotrophs". Bacteria and archaea in the soil and in the roots of some plants have the ability to convert molecular nitrogen from the air (N_2) to ammonia (NH_3), thereby breaking the tough triple bond of molecular nitrogen. From here, various microorganisms convert ammonia to other nitrogen compounds that are easier for plants to use. **In this way, plants get their nitrogen indirectly from the air via microorganisms in the soil and in certain plant roots.**

X-Factor:

- Nitrogen is a key component of amino acids, which form the building blocks of plant proteins and enzymes. Proteins make up the structural materials of all living matters and enzymes facilitate the vast array of biochemical reactions within a plant.
- Nitrogen is also a component of the chlorophyll molecule, which enables the plant to capture sunlight energy by photosynthesis, driving plant growth and grain yield.

Source: <https://www.thehindu.com/sci-tech/science/new-pathway-to-regulate-nitrate-absorption-in-plants/article65590908.ece>

https://kochagronomicservices.com/knowledge-center/The-Role-of-Nitrogen-in-Crop-Production-and-How-to-Protect-It_2288.aspx

<https://www.agqlabs.us.com/macronutrients-in-plants/>

<https://www.wtamu.edu/~cbaird/sq/2014/02/21/how-do-plants-get-their-nitrogen-from-the-air/#:~:text=Such%20organisms%20are%20called%20%22diazotrophs,and%20in%20certain%20plant%20roots.>

86. (a)

Explanation:

Statement 1 is incorrect: As part of a global 'China-plus-one' strategy adopted post the Covid-19 pandemic that caused massive supply-chain disruptions, **a group of western nations are cooperating** to develop alternatives to China to ensure key industrial supplies.

Statement 2 is correct: The US and 10 partners — Australia, Canada, Finland, France, Germany, Japan, the Republic of Korea (South Korea), Sweden, the United Kingdom, and the European Commission — have come together to form the MSP. The new grouping **is aimed at catalysing investment from governments and the private sector to develop strategic opportunities and to bolster critical mineral supply chains.**

Statement 3 is incorrect: India is not part of this arrangement — called the Minerals Security Partnership (MSP) — **but New Delhi is working through diplomatic channels to fetch an entry.**

X-Factor:

- The 17 rare earth elements (REE) include the 15 Lanthanides (atomic numbers 57 — which is Lanthanum — to 71 in the periodic table) plus Scandium (atomic number 21) and Yttrium (39). REEs are classified as light RE elements (LREE) and heavy RE elements (HREE) .
- Rare-earth elements (REE) are necessary components of more than 200 products across a wide range of applications, especially high-tech consumer products, such as cellular telephones, computer hard drives, electric and hybrid vehicles, and flat-screen monitors and televisions. Significant defense applications include electronic displays, guidance systems, lasers, and radar and sonar systems.
- Some REEs are **available in India** — such as **Lanthanum, Cerium, Neodymium, Praseodymium and Samarium**, etc. Others such as **Dysprosium, Terbium, and Europium**, which are classified as HREEs, are not available in Indian deposits in extractable quantities. Hence, there is a dependence on countries such as **China** for HREEs, which is one of the leading producers of

REEs, with an estimated **70 per cent share of the global production.**

Source:

<https://indianexpress.com/article/explained/explained-sci-tech/explained-rare-earth-elements-india-global-alliance-supply-8070563/>

<https://www.state.gov/minerals-security-partnership-convening-supports-robust-supply-chains-for-clean-energy-technologies/>

87. (c)

Explanation:

Option (c) is correct:

Rainwater harvesting is a simple strategy by which rainfall is gathered and stored for future usage. The process involves collection and storage of rainwater with help of artificially designed systems, that runs off natural or man-made catchment areas e.g. rooftop, compounds, rocky surface, hill slopes or artificially repaired impervious/semi-pervious land surface. The collected rainwater from surfaces on which rain falls may be filtered, stored and utilized in different ways or directly used for recharge purposes.

- **Paar system:** Paar is a common water harvesting practice in the western Rajasthan region. It is a common place where the rainwater flows from the agar (catchment) and in the process percolates into the sandy soil. The structure was constructed through traditional masonry technology. Rainwater harvested through PAAR technique is known as Patali pani.
- **Bandhis:** Bandhis/ Talabs are reservoirs. They may be natural, such as the ponds (pokhariyan) at Tikamgarh in the Bundelkhand region. They can be human-made, such the lakes in Udaipur. A reservoir area of less than five bighas is called a talai ; a medium sized lake is called a bandhi or talab; bigger lakes are called sagar or samand. The pokhariyan serve irrigation and drinking purposes. When the water in these reservoirs dries up just a few days after the monsoon, the pond beds are cultivated with rice.
- **Pat system:** Bhitada village, Jhabua district of Madhya Pradesh developed the

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unique pat system. This system was devised according to the peculiarities of the terrain to divert water from swift-flowing hill streams into irrigation channels called pats.

- **Johad:** Johads are small earthen check dams that capture and conserve rainwater, improving percolation and groundwater recharge. Starting 1984, the last sixteen years have seen the revival of some 3000 johads spread across more than 650 villages in Alwar district, Rajasthan. This has resulted in a general rise of the groundwater level by almost 6 metres and a 33 percent increase in the forest cover in the area. Five rivers that used to go dry immediately following the monsoon have now become perennial, such as the River Arvari, has come alive.

X-Factor:

Examples of Traditional water harvesting system in India

- **Trans-Himalayan Region**
Zing -Tanks for collecting water from melted ice in Ladakh.
- **Western Himalayas**
Kul -Water channels in mountain areas of Jammu, Himachal Pradesh.
Naula -Small ponds in Uttaranchal.
- **Eastern Himalayas**
Northeastern Hill Ranges
Apatani system -Terraced plots connected by inlet and outlet channels in Arunachal Pradesh.
Zabo -Impounding runoff in Nagaland
Bamboo drip irrigation – Water from streams in the hills is brought to the plains via bamboo pipes for drip irrigation in Meghalaya
- **Brahmaputra Valley**
Dongs – Ponds in Assam
Dungs or Jampoos – small irrigation channels linking rice fields to streams in the Jalpaiguri district of West Bengal
- **Indo-Gangetic Plain**
Dighis -Small square or circular reservoir fed by canals from rivers in Delhi
Baolis – secular structured stepwells from which everyone could draw water and use for washing and bathing.
- **Thar Desert**
Nadis / Talab

Baoris / Bers -Community wells in Rajasthan

Tankas -Underground tank Bikaner in Rajasthan

Kund – a circular underground well; having a saucer-shaped catchment area that gently slopes towards the center where the well is situated.

Source:

<https://www.cseindia.org/traditional-water-harvesting-systems-683>

<https://blog.mygov.in/water-conservation-rainwater-harvesting/>

<https://www.thehindu.com/news/national/other-states/special-story-monsoon-rains-fill-up-traditional-water-harvesting-structures/article65737048.ece>

88. (a)

Explanation:

Option(a) is **correct:**

The committee is appointed under retired Supreme Court Judge Justice BN Srikrishna to report on "Data Protection Framework" to the Government. The main recommendation given by the committee are:

- Personal data shall be processed only for purposes that are clear specific and lawful.
- All firms and agencies have to appoint data protection officers.
- The data protection officers shall act as a point of contact for individual.
- The centre shall notify data protection authority of India.
- Individual will have the right to withdraw consent.
- Firms will have to ensure at least one copy of personal data to be stored in India.
- Critical personal data shall only be processed in a server or data center located in India.

X-Factor:

Justice K.S. Puttaswamy (retd.) vs Union of India:

The judgment delivered on that date formally recognised the right to privacy as being a fundamental right stemming from the right to life and personal liberty, guaranteed under Article 21 of the Indian Constitution.

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The Bench also held that while the right to privacy is intrinsic to an individual's ability to exercise bodily autonomy, it is still not an "absolute right" in and of itself, placing limitations in a manner similar to those placed on the right to free speech and expression.

Source:

<https://indianexpress.com/article/explained/explained-sci-tech/personal-data-protection-bill-withdrawal-reason-impact-explained-8070495/>

<https://economictimes.indiatimes.com/news/politics-and-nation/justice-bn-srikrishna-committee-submits-report-on-data-protection-herere-the-highlights/articleshow/65164663.cms?from=mdr>

<https://globalfreedomofexpression.columbia.edu/cases/puttaswamy-v-india/>

89. (a)

Explanation:

Option (a) is correct: *Kurxinji, Mullai, Marutham, Neithal and Paalai are five ecological regions mentioned in ancient Tamil Sangam literature.* Soil samples from these ecological regions mentioned in ancient Tamil Sangam literature have been collected and sent to the capital which will be featured in a project relating to the new Parliament.

X-Factor:

Sangam literature is the earliest writings in the Tamil language, thought to have been produced in three *chankams*, or literary academies, in Madurai, from the 1st to the 4th century. The *Tolkappiyam*, a book of grammar and rhetoric, and eight anthologies (*Ettuttokai*) of poetry were compiled—*Ainkurunuru*, *Kuruntokai*, *Narrinai*, *Akananuru*, *Kalittokai*, *Patirruppattu*, *Purananuru*, and *Paripatal*. A ninth anthology, *Pattupattu*, consists of 10 idylls that present a picture of early Tamil life.

Source: <https://www.thehindu.com/news/national/tamil-nadu/tn-sends-soil-samples-for-project-relating-to-new-parliament-building/article65599930.ece>
<https://www.britannica.com/art/Nayana>

90. (d)

Explanation:

Statement 1 is correct: **Santhal**, also called **Manjhi**, ethnic group of eastern India, numbering well over five million at the turn of the 21st century. Their greatest concentration is in the states of Bihar, Jharkhand, West Bengal, and Orissa, in the eastern part of the country. Some 200,000 also live in Bangladesh and more than 10,000 in Nepal

Statement 2 and 3 are correct: The Santhal have 12 clans, each divided into a number of subdivisions also based on descent, **which is patrilineal**. Traditionally, members of the same clan do not marry each other. Membership in the clan and subclan carries certain injunctions and prohibitions with regard to style of ornament, food, housing, and religious ritual. Marriage is generally monogamous; polygyny, though permitted, is rare. The traditional religion centres on the worship of spirits, and the ancestral spirits of the headmen are objects of an important cult.

Statement 4 is correct: Born in 1958 in Baidaposi village of Mayurbhanj district, Ms President Murmu belongs to the Santhal community, one of India's largest tribal groups.

X-Factor:

- The language of Santhal tribes is Santhali, a dialect of Kherwari, a Munda (Austroasiatic) language.

Source:

<https://indianexpress.com/article/explained/droupadi-murmu-tribal-revolts-president-murmu-inaugural-speech-8051090/>

<https://www.britannica.com/topic/Santhals>

<https://www.bbc.com/news/world-asia-india-61892776>

91. (d)

Explanation:

Option (d) is correct:

UJJAWALA: A Comprehensive Scheme for Prevention of trafficking and Rescue, Rehabilitation and Re-integration of Victims of Trafficking and Commercial Sexual Exploitation.

Objective of the scheme:

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- To **prevent trafficking of women and children** for commercial sexual exploitation through **social mobilization and involvement of local communities, awareness** generation programmes, generate public discourse through workshops/seminars and such events and any other innovative activity.
- To **facilitate rescue of victims** from the place of their exploitation and place them in safe custody.
- To **provide rehabilitation services** both immediate and long-term to the victims by providing basic amenities/needs such as shelter, food, clothing, medical treatment including counselling, legal aid and guidance and vocational training.
- To **facilitate reintegration of the victims** into the family and society at large
- To **facilitate repatriation of cross-border victims** to their country of origin.

X-Factor:

Pradhan Mantri Ujjwala Yojana (PMUY)

- It aims to safeguard the health of women & children by providing them with a clean cooking fuel – LPG, so that they don't have to compromise their health in smoky kitchens or wander in unsafe areas collecting firewood.
- Pradhan Mantri Ujjwala Yojana was launched in Ballia.
- Ensuring women's empowerment, especially in rural India, the connections will be issued in the name of women of the households.
- Identification of the BPL families will be done through Socio Economic Caste Census Data.

Source:

<https://wcd.nic.in/schemes/ujjawala-comprehensive-scheme-prevention-trafficking-and-rescue-rehabilitation-and-re>

<https://womenandchildren.assam.gov.in/information-services/ujjawala-new-scheme-for-woman>

<https://web.umang.gov.in/landing/departments/pradhan-mantri-ujjawala-yojna-pmuy.html>

92. (a)

Explanation:

India's mineral and mining sector operates under a federal structure wherein the **Central Government formulates the legislation for all minerals except the minor minerals and the State Governments formulate legislation for minerals classified as minor minerals.** India has written legal and constitutional framework to manage the mineral sector. National Mineral Policy provides the direction for mineral sector. **Management of mining sector is the responsibility of the Central Government and the State Governments.** The Constitution bestows power to the **Parliament to enact legislation relating to the mining and the States are bound by the Central legislation.** In the federal structure of India, the State Governments are the owners of minerals located within their respective boundaries. The Central Government is the owner of the minerals underlying the ocean within the territorial waters or the Exclusive Economic Zone of India.

Statement 1 is correct: Minor minerals have been **defined in Section 3 of the mines and Minerals (Regulation and Development) Act, 1957** and comprise mostly such minerals that are locally available and have local use.

Statement 2 is incorrect: The Ministry of Mines (MoM), Government of India is responsible for the entire minerals and mining sector in the country that includes legislation, administration, policy formulation etc. in respect of all mines and minerals other than coal and lignite, natural gas and petroleum, but including offshore minerals. In the case of atomic minerals and coal activities of the Ministry are limited to regional exploration. **In India, the minerals are classified as minor minerals and major minerals. The power to frame policy and legislation relating to minor minerals is entirely delegated to the State Governments while policy and legislation relating to the major minerals is dealt by the Ministry of Mines.**

The Ministry is responsible for the administration of the Mines and Minerals (Development and Regulation) Act, 1957 and rules made there under in respect of all

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mines and minerals other than coal, natural gas and petroleum.

Sources:

<https://dgmup.gov.in/en/page/minor-minerals-list>

<https://mines.gov.in/writereaddata/UploadFile/ebookmines.pdf#:~:text=In%20India%2C%20the%20minerals%20are%20classified%20as%20minor,the%20major%20minerals%20is%20dealt%20by%20the%20MoM.>

93. (d)

Explanation:

A landslide is defined as the movement of a **mass of rock, debris, or earth down a slope. Landslides are a type of "mass wasting,"** which denotes any down-slope movement of soil and rock **under the direct influence of gravity.**

Statement 1 is correct: In mountainous regions, landslides can occur due to **snowmelt, falling of rocks and debris.** Extreme ice-snow melting in winter affects the infiltration process of snow water on the slope surface significantly and plays an important role in **the deformation stability of landslide.**

Statement 2 and 3 are correct: **All landslides in the hilly states of North East India have been caused due to weak lithology,** intense fracturing of rocks, steep slopes, **thick mantle of waste and excessive rainfall in short duration (Flash flood).**

X Factor:

The term "landslide" encompasses five modes of slope movement: falls, topples, slides, spreads, and flows. These are further subdivided by the type of geologic material (bedrock, debris, or earth). Debris flows (commonly referred to as mudflows or mudslides) and rock falls are examples of common landslide types.

Source:

<https://www.firstpost.com/india/explained-why-landslides-are-so-frequent-in-the-monsoon-10881621.html>

<https://www.hindawi.com/journals/ace/2020/8832485/>

94. (c)

Explanation:

Option(c) is correct:

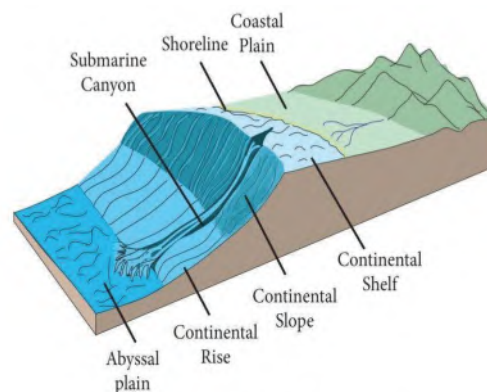


Figure 5.8 Ocean relief

The bottom of the ocean has a variety of landforms just as it is seen on the earth's surface. There are large mountain ridges, deep depressions, flat plains, basins and volcanoes. The configuration of an ocean floor is shown with the help of a 'Hypsometric curve' or 'Hypsographic curve'. It is a graph denoting the proportion of a landmass standing above or below the sea level.

Ocean relief features (from deep sea to shoreline):

1. Abyssal plain
2. Continental Rise
3. Continental Slope
4. Continental Shelf.

X factor:

1. The Abyssal plain is the vast area of flat terrain in the bottom of the oceans. It is the largest part of ocean relief covering more than 50% of the total area.
2. There is an accumulation of very fine sediments on the floor. The sediments are combinations of fine particles of clay and microorganisms.
3. As in the case of sedimentary rocks of earth's surface these sediments are in layers and are used to trace geological events in the past.

Source:

SCERT: TN: XI: Geography: Chapter: 5: Hydrosphere: Page:113-115.

95. (d)

Explanation:

Statement 1 and 2 are correct: The Human Development Report 2021/2022,

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compiled by the United Nations Development Programme, has put India in the 132nd position out of 191 countries. India ranked 130 among 189 countries in 2020.

Statement 3 is correct: The current HDI value of the country is 0.633, which is significantly below the global average of 0.732. As if that is not bad enough, India is ranked behind the world average on all the four parameters: **Bangladesh, Bhutan, China, and Sri Lanka have performed better.**

X-factor:

- Admittedly, India's showing is consistent with the global trend: 90 per cent of the countries surveyed have recorded a fall on the index. In fact, this is the first time since 1990 — when the HDI was constituted — that the global HDI value has decreased for two years in a row owing to factors such as the Covid-19 pandemic and the Ukraine war.
- This dip leads to two concerns: first, the momentum gained in realising sustainable development goals has suffered reversals; second, the report mentions that economies are facing a “new uncertainty complex” as a result of intersecting factors like widening inequalities and a polarising political ethic.

Source:

<https://www.telegraphindia.com/opinion/backbencher-editorial-on-indias-dismal-performance-in-human-development-index/cid/1887688>

96. (c)

Explanation:

Option(c) is correct:

Temperate grasslands are called differently in different parts of the world.

Prairies	-- North America
Steppes	-- Eurasia
Pampas	-- Argentina and Uruguay
Veld	-- South Africa
Downs	-- Australia
Canterburg	-- Newzealand
Manchurian	-- China

- Temperate Grasslands are usually found in the **interior of the continents** and are characterized by large seasonal temperature variations, with warm summer and cold winter.
- The type of grassland in these regions **strongly depends upon precipitation.**
- Higher precipitation leads to tall and soft grass and lower precipitation leads to short and soft grass.

X factor:

These regions favor wheat cultivation. Extensive mechanized agriculture is practiced due to lack of farm labour. Pastoral industry becomes the main occupation, thereby facilitating slaughtering of animals, packing of raw and processed meat, dairy products etc.

Source: SCERT: TN: XI: Geography: Chapter 3: Lithosphere Endogenic process: Page 53, 54.

97. (b)

Explanation:

Scientists have developed a sustainable process to make ‘space bricks’ that would allow infrastructure construction on the Moon and even Mars. The process will allow the construction **of habitats on the Moon and eventually on Mars.** In a special report, IISc revealed that the method uses bacteria and guar beans and exploits the lunar and Martian soil to consolidate the content into tough, load-bearing brick-like structures.

Statement 1 is incorrect: Scientists from the **Indian Space Research Organisation (ISRO) and Bengaluru’s Indian Institute of Science (IISc)** have developed a sustainable process to make ‘space bricks’ together.

Statement 2 is correct: The method for making these space bricks has been outlined in a study published in PLOS One. A slurry is first created by mixing Martian soil with guar gum, **a bacterium called sporosarcina pasteurii**, urea and nickel chloride (NiCl₂). This slurry can be poured into moulds of any desired shape, and over a few days the bacteria convert the urea into crystals of calcium carbonate. These crystals, along with biopolymers secreted by the microbes, act as cement holding the soil particles together.

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Statement 3 is correct: In space brick manufacturing the raw material required for manufacturing the bricks are **urea, which can be sourced from human urine and lunar soil.**

X Factor:

MARS (Martian Atmosphere Simulator)

It can create atmospheric conditions similar to Mars inside a lab.

The scientists have also created a lab-on-a-chip device that aims to measure bacterial activity in micro-gravity conditions.

These devices will soon be sent into space with ISRO's help to study bacterial growth in low gravity.

Source:

<https://www.republicworld.com/science/space/isro-iisc-develop-space-bricks-for-habitat-construction-on-moon-and-mars-articleshow.html>

<https://indianexpress.com/article/cities/bangalore/bengaluru-iisc-researchers-bricks-martian-soil-bacteria-urea-7879110/>

98. (d)

Explanation:

The term '**drought**' is applied to an extended period when there is a shortage of water availability due to inadequate precipitation, excessive rate of evaporation and over-utilisation of water from the reservoirs and other storages, including the ground water.

Statement 1 is incorrect: Meteorological Drought: It is a situation when there is a prolonged period of inadequate rainfall marked with mal-distribution of the same over time and space.

Hydrological Drought: It results when the availability of water in different storages and reservoirs like aquifers, lakes, reservoirs, etc. falls below what the precipitation can replenish.

Statement 2 is incorrect: Agricultural Drought: It is also known as **soil moisture drought, characterised by low soil moisture** that is necessary to support the crops, thereby resulting in crop failures. Moreover, if an area has more than 30 per cent of its gross cropped area under irrigation, the area is excluded from the drought-prone category.

X-Factor: Drought prone areas in India:

Extreme Drought Affected Areas: Most parts of Rajasthan, particularly areas to the west of the Aravali hills, i.e., Marusthali and Kachchh regions of Gujarat fall in this category. Included here are also the districts like Jaisalmer and Barmer from the Indian desert that receive less than 90 mm average annual rainfall.

Severe Drought Prone Area: Parts of eastern Rajasthan, most parts of Madhya Pradesh, eastern parts of Maharashtra, interior parts of Andhra Pradesh and Karnataka Plateau, northern parts of interior Tamil Nadu and southern parts of Jharkhand and interior Odisha are included in this category.

Moderate Drought Affected Area: Northern parts of Rajasthan, Haryana, southern districts of Uttar Pradesh, the remaining parts of Gujarat, Maharashtra except Konkan, Jharkhand and Coimbatore plateau of Tamil Nadu and interior Karnataka are included in this category. The remaining parts of India can be considered either free or less prone to the drought.

Source: 11th NCERT India Physical Environment, Chapter 7, Page no. 88, 89, 90

99. (b)

Explanation:

TULIP Program would help **enhance the value-to-market of India's graduates and help create a potential talent pool in diverse fields like urban planning, transport engineering, environment, municipal finance etc.** thus not only catalyzing creation of prospective city managers but also talented private/ non-government sector professionals. **It would benefit ULBs and smart cities immensely.** It will lead **to infusion of fresh ideas and energy with engagement of youth in co-creation of solutions** for solving India's urban challenges. More importantly, it will further Government's endeavors to boost community partnership and government-academia-industry-civil society linkages

Statement 1 is incorrect: The Urban learning Internship Program (TULIP), was launched in June 2020. This venture is jointly launched by **the Ministry of Human**

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Resource Development (Ministry of Education), the Ministry of Housing & Urban Affairs, and the All-India Council for Technical Education (AICTE).

Statement 2 is correct: It is an online portal for 'The Urban Learning Internship Program (TULIP)' - A program for **providing internship opportunities to fresh graduates in all Urban Local Bodies (ULBs) and Smart Cities** across the country.

Source:

<https://pib.gov.in/PressReleasePage.aspx?PRID=1629314>

https://internship.aicte-india.org/module_ulb/Dashboard/TulipMain/

<https://www.india.com/education/govt-internships-what-is-tulip-govts-urban-learning-internship-program-meant-to-empower-fresh-graduates-5400972/>

100. (a)

Explanation:

Option (a) is correct:

Thus, a mineral is a **naturally occurring organic and inorganic substance**, having an orderly atomic structure and a **definite chemical composition and physical properties**. A mineral is composed of two or more elements. **But, sometimes single element minerals like sulphur, copper, silver, gold, graphite etc. are found.**

Metallic Minerals:

These minerals contain metal content and can be sub-divided into three types: (i) Precious metals: gold, silver, platinum etc. (ii) Ferrous metals: iron and other metals often mixed with iron to form various kinds of steel. (iii) Non-ferrous metals: include metals like copper, lead, zinc, tin, aluminums etc.

Non-Metallic Minerals:

These minerals do not contain metal content. Sulphur, phosphates and nitrates are examples of non-metallic minerals. Cement is a mixture of non-metallic minerals.

Source: NCERT, Physical geography Page no. 44